Robust Tensor Low-Rank Sparse Representation With Saliency Prior for Hyperspectral Anomaly Detection

Qingjiang Xiao, *Student Member, IEEE*, Liaoying Zhao, Shuhan Chen, *Member, IEEE*, and Xiaorun Li

***Abstract*— Recently, hyperspectral anomaly detection (HAD) methods based on tensor low-rank representation (TLRR) have received widespread attention. However, most of them tend to emphasize the utilization of multiple types of prior knowledge to characterize background components, while the prior information about anomaly components is limited. In addition, the constructed background dictionary is also susceptible to noise and outliers. To address these challenges, this article focuses on both the background and abnormal components, proposing a robust tensor low-rank sparse representation with saliency prior (RTLSR-SP) method for HAD. Specifically, for the background component described by the dictionary tensor and the corresponding coefficient tensor, tensor nuclear norm (TNN) constraint and sparsity constraint are imposed on the coefficient tensor simultaneously to capture the global and local spatial–spectral structure information of the hyperspectral image (HSI), respectively. For the anomalous component, we design a sparse saliency prior weight tensor to enhance the saliency of anomalous targets. Meanwhile, the tensor $\ell_{F,1}$-norm is also integrated into the model to better separate abnormal targets from the background. Furthermore, combining tensor robust principal component analysis (TRPCA) and skinny tensor singular value decomposition (skinny t-SVD), a robust background dictionary is constructed. Finally, an efficient iterative algorithm based on the alternating direction method of multipliers (ADMM) is derived to optimize the RTLSR-SP model. Comprehensive experimental findings on one simulated dataset and six real hyperspectral datasets demonstrate the effectiveness and superiority of the proposed algorithm compared with eight state-of-the-art HAD algorithms.**

***Index Terms*— Anomaly detection, hyperspectral image (HSI), sparse saliency prior weight tensor, tensor low-rank sparse representation (SR), tensor robust principal component analysis (TRPCA).**

NOMENCLATURE

a	Scalar.
$\boldsymbol{A}$	Matrix.
$\boldsymbol{A}_{ij}$ or a_{ij}	(i, j)th entry of $\boldsymbol{A}$.
$\boldsymbol{\mathcal{A}}(:,:,k)$ or $\boldsymbol{A}^{(k)}$	kth frontal slice of $\boldsymbol{\mathcal{A}}$.
$(\cdot)^*$	Transpose operator.
$\|\boldsymbol{a}\|_2 = (\sum_i \|a_i\|^2)^{1/2}$	ℓ_2-norm of $\boldsymbol{a}$.
$\|\boldsymbol{A}\|_1 = \sum_{i,j} \|a_{i,j}\|$	ℓ_1-norm of $\boldsymbol{A}$.
$\|\boldsymbol{A}\|_F = (\sum_{ij} \|a_{ij}\|^2)^{1/2}$	Frobenius norm of $\boldsymbol{A}$.
$\boldsymbol{A}_{(i)}$	Mode-i matricization of $\boldsymbol{\mathcal{A}}$.
$\sigma_i(\boldsymbol{A})$	Singular values of $\boldsymbol{A}$.
$\mathrm{Tr}(\boldsymbol{A})$	Trace of $\boldsymbol{A}$.
$< \boldsymbol{A}, \boldsymbol{B} >= \mathrm{Tr}(\boldsymbol{A}^T \boldsymbol{B})$	Inner product of $\boldsymbol{A}$ and $\boldsymbol{B}$.
$\boldsymbol{a}$	Vector.
$\boldsymbol{\mathcal{A}}$	Tensor.
$\boldsymbol{\mathcal{A}}_{ijk}$ or a_{ijk}	(i, j, k)th entry of $\boldsymbol{\mathcal{A}}$.
$\boldsymbol{\mathcal{A}}^{-1}$	Inverse of $\boldsymbol{\mathcal{A}}$.
$\|\cdot\|$	Absolute value.
$\|\boldsymbol{A}\|_{2,1} = \sum_j \|\boldsymbol{A}(:, j)\|_2$	$\ell_{2,1}$-norm of $\boldsymbol{A}$.
$\|\boldsymbol{\mathcal{A}}\|_1 = \sum_{ijk} \|a_{ijk}\|$	ℓ_1-norm of $\boldsymbol{\mathcal{A}}$.
$\|\boldsymbol{\mathcal{A}}\|_F = (\sum_{ijk} \|a_{ijk}\|^2)^{1/2}$	Frobenius norm of $\boldsymbol{\mathcal{A}}$.
$\|\boldsymbol{\mathcal{A}}\|_\infty = \max_{ijk} \|a_{ijk}\|$	Infinity norm of $\boldsymbol{\mathcal{A}}$.
$\|\boldsymbol{A}\|_* = \sum_i \sigma_i(\boldsymbol{A})$	Nuclear norm of $\boldsymbol{A}$.
$\mathrm{rank}(\boldsymbol{A})$	Rank of $\boldsymbol{A}$.
$< \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{B}} >= \sum_{i=1}^{n_3} < \boldsymbol{A}^{(i)}, \boldsymbol{B}^{(i)} >$	Inner product of $\boldsymbol{\mathcal{A}}$ and $\boldsymbol{\mathcal{B}}$.

Manuscript received 10 July 2023; revised 28 September 2023; accepted 30 October 2023. Date of publication 2 November 2023; date of current version 10 November 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62171404 and in part by the Joint Fund Project of the Chinese Ministry of Education under Grant 8091B022120. *(Corresponding author: Liaoying Zhao.)*

Qingjiang Xiao and Liaoying Zhao are with the School of Computer Science and Technology, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: qingjiangxiao@foxmail.com; zhaoly@hdu.edu.cn).

Shuhan Chen and Xiaorun Li are with the Department of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: 11410057@zju.edu.cn; lxr@zju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3329510

I. INTRODUCTION

HYPERSPECTRAL image (HSI) is a 3-D data cube that contains 2-D spatial information used to describe the distribution of ground objects and 1-D spectral information that reflects the intrinsic properties of materials and has been widely used in various fields, such as anomaly detection, target detection, and classification [1], [2], [3], [4]. Among them, anomaly detection is an important technique for detecting targets of interest in HSIs, which can be used in military and civilian fields, such as camouflage recognition, disaster monitoring, and precision agriculture [5]. In the past decades, scholars have done a lot of work in the field of hyperspectral anomaly detection (HAD), resulting in various anomaly detection algorithms. According to the specific theoretical knowledge adopted, existing anomaly detection algorithms can be roughly divided into three categories: statistics-based methods [6], [7], [8], [9], component decomposition-based methods [10], [11], [12], [13], [14], [15], [16], and representation-based methods [17], [18], [19], [20], [21]. In this article, we mainly study representation-based methods.

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

Representation-based methods have received widespread attention because they do not require any specific assumptions about the background distribution. The core idea is to use a representation coefficient and a specific dictionary to reconstruct all HSI pixels, where background pixels can be represented by several atoms in the overcomplete dictionary, while abnormal pixels cannot or have a relatively large reconstruction error. The reconstruction error represents the abnormality degree of the pixel. The larger the error, the higher the probability that the pixel becomes an abnormal target. Moreover, according to different regularization constraints imposed on the representation coefficient, representation-based methods can be subdivided into sparse representation (SR)-based methods [22], [23], collaborative representation (CR)-based methods [24], [25], [26], low-rank representation (LRR)-based methods [27], [28], [29], [30], and tensor LRR (TLRR)-based methods [31], [32], [33], [34], [35]. In fact, the first three categories of approaches are all dependent on matrix operations, which inevitably devastate the spatial–spectral structure of the original HSI data, resulting in a decrease in detection performance.

As an effective high-dimensional data analysis tool, tensor representation can describe multiple characteristic properties of HSI through high-order structure and can better retain the spatial–spectral information of HSI. In recent years, some HAD approaches based on tensor representation have been proposed [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. Although some existing HAD approaches based on tensor representation have reached initially satisfactory detection results, there are still some problems to be solved.

1) The construction of the background dictionary is susceptible to noise and outliers contamination. Abnormal targets with low signal-to-noise ratio are frequently submerged in the background, causing the background dictionary to not fully represent the background materials, thus affecting subsequent detection performance.
2) Most methods only focus on the global structural information of background components and ignore the mining of local structural information. When the target belongs to high-frequency information, insufficient mining of background components may lead to the omission of small target detailed features.
3) Most methods only impose some kind of sparsity norm on the abnormal component to improve the group sparsity of abnormal pixels. However, other prior information on the abnormal component such as structural sparsity is not fully exploited, which may not be able to distinguish anomalies from the background well. Subsequently, these three issues will be discussed separately.

The quality of the background dictionary plays a crucial role in TLRR-based HAD methods. One of the simplest background dictionary construction methods is to directly use the original HSI tensor data as a background dictionary under the assumption that noise and outliers are ignored, such as [32], [33], and [35]. However, in practical applications, due to the limitations of various objective factors, such as illumination conditions, transmission conditions, and imaging instruments, HSI data will inevitably be polluted by various noises during the acquisition and transmission process. In addition, the computational complexity is significantly increased due to the vast number of atoms present in such a dictionary. To decrease the atom count and choose representative atoms, Sun et al. [34] designed a dictionary construction approach utilizing simple linear iterative clustering (SLIC) for generating a background dictionary matrix. Due to the use of matrix operations in the construction process, the inherent structural information of HSI tensor data is destroyed. Considering the high computational complexity and noise interference, Wang et al. [31] first used PCA technology to preprocess the original HSI tensor data and then used the tensor robust principal component analysis (TRPCA) [41] method to construct a clean background dictionary tensor. However, PCA is sensitive to small abnormal targets, and using PCA preprocessing operations may lead to a higher false alarm rate. Based on the above analysis, it is necessary to construct a robust and efficient background dictionary. In this article, we combine TRPCA and skinny tensor singular value decomposition (skinny t-SVD) to construct a background dictionary tensor. Meanwhile, an adaptive approach is devised to estimate the tensor tube rank with the aim of comprehensively characterizing the background and eliminating anomalies.

HAD can be regarded as a binary classification problem in essence, and the use of prior knowledge of background components is the key technology to solve this kind of problem, among which the most commonly used priors are low-rank and piecewise smooth properties. For example, Xu et al. [15], Wang et al. [31], He et al. [32], Wang et al. [33], and Ruhan et al. [37] focus on mining the low-rank characteristic of the background in order to accurately estimate the rank of the background. Li et al. [14], Sun et al. [34], Li et al. [36], Wang et al. [39], and Feng et al. [40] simultaneously mine the low-rank and piecewise smoothing characteristics of the background component. In addition, some scholars have also studied the global structure (low-rank constraint) and local structure (sparse constraint) information of the background component, such as [18] and [42]. By simultaneously imposing low-rank and sparse constraints on the coefficient matrix, the learned coefficient matrix can achieve more precise subspace representation capabilities. However, in [18] and [42], only 2-D data are supported for direct processing. For 3-D HSI, it is necessary to convert it into 2-D HSI, which may potentially compromise the multidimensional structural information of the HSI tensor data. To solve this problem, we impose both low-rank constraint and sparse constraint on the coefficient tensor in the tensor domain to capture the global and local spatial–spectral structure information of the background component in HSI tensor data, respectively. Each lateral slice in the coefficient tensor is an approximate representation of the original HSI data under the offered background dictionary tensor. The main discovery is that an SR of a data sample ideally corresponds to a combination of several atoms in its own subspace. The sparsity constraint automatically selects some atoms from a given data sample, which are consistently in close proximity to it and belong to the same subspace. Consequently, the tubes corresponding to lateral slices in the coefficient tensor belonging to the same subspace exhibit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

higher values, whereas the tubes corresponding to lateral slices belonging to other subspaces tend to approach or reach zero [43], [44].

Abnormal targets in HSIs have the characteristics of low probability of occurrence, small size, unbalanced distribution, and so on. Most of the existing HAD methods impose an ℓ-norm constraint on the abnormal matrix or abnormal tensor to promote the group sparsity of abnormal pixels, such as matrix $\ell_{2,1}$-norm [10], [11], [18], [19] and tensor $\ell_{2,1}$-norm [32], [34], [39]. These proposed norms may not fully utilize the prior information of anomalous components. Based on the structure tensor theory, Luo et al. [45] combined the image entropy operator and the prior information filter to design a sparse prior map for infrared small target detection and achieved satisfactory performance. Inspired by this, we create an adaptive weight map and a novel iterative sparse reweighting weight map by merging the spectral characteristics of HSI with the sparse characteristics of anomalous targets. Finally, we generate a new sparse saliency prior weight map for weighting the anomaly tensor in the hopes of enhancing the saliency of anomaly targets while suppressing the background.

In view of the above analysis, in order to address these three problems simultaneously, this article makes full use of the prior information on the background and abnormal components and proposes a robust tensor low-rank SR with saliency prior (RTLSR-SP) model for HAD. As a conclusion, the main innovative contributions of this article can be depicted as follows.

1) We develop a TLRR-based RTLSR-SP model for the HAD task. The model simultaneously considers the global low-rank property and local similarity of the background, the structural sparsity of anomalies, and the robustness of the background dictionary. Moreover, an efficient iterative algorithm based on the alternating direction method of multipliers (ADMM) is proposed to optimize the RTLSR-SP model.
2) Tensor nuclear norm (TNN) constraint and sparsity constraint are applied to the coefficient tensor to capture the global and local spatial–spectral structure information of the background component in the HSI tensor data, respectively.
3) To enhance the saliency of anomalous targets, we design a sparse saliency prior weight tensor to weight the anomaly tensor. Besides, the tensor $\ell_{F,1}$-norm is incorporated into our approach to better separate anomalous targets from the background.
4) We design an adaptive method to estimate tensor tube rank and construct a robust and efficient background dictionary tensor. The constructed background dictionary tensor is robust to noise and can fully represent the background.

The remainder of this article is arranged as follows. Some related work is presented in Section II. In Section III, we introduce the proposed RTLSR-SP model and the optimization process in detail. The experimental results and analysis can be found in Section IV. Finally, Section V concludes this article.

II. Related Work

In this section, we introduce some related algorithms of HAD, mainly including matrix-based representation methods and tensor-based representation methods.

A. Matrix-Based Representation Methods

The most representative anomaly detector based on SR is the BJSR algorithm of background joint SR proposed by Li et al. [22], which constructs the background by adaptively searching representative atoms in local regions. In [23], a sparse score estimation framework is designed to enhance the diversity between background pixels and abnormal pixels in the learned dictionary. However, the construction of an overcomplete dictionary free from abnormal pixel pollution remains challenging in the absence of prior information. Li and Du [24] put forward a seminal approach known as the CR detector (CRD), which postulates that background pixels can be linearly represented by pixels located between outer and inner windows, while abnormal pixels cannot. Unfortunately, when the sizes of the inner and outer windows are set unreasonably, the constructed background dictionary is easily contaminated by abnormal pixels, leading to unsatisfactory detection results. To mitigate this phenomenon, a great deal of variants, including CRD-based background purification framework [25], dual CRD [26], self-weighted CRD detector [46], and relaxed CRD detector [17], are studied to improve anomaly detection performance. CR-based methods only consider local characteristics when modeling background and fail to effectively utilize global characteristics, resulting in limited anomaly detection performance.

Unlike SR and CR, LRR-based detectors aim to find the lowest rank representation of all pixels. For instance, in [18], a low-rank and SR (LRASR) algorithm is proposed for HAD. LRASR constructs a cluster-based dictionary to describe the background and obtains the coefficient matrix through low-rank and sparse constraints to explore global and local structural information. By combining graph and total variation (TV) regularization into the LRR model, Chen and Wang [19] explored a GTVLRR algorithm to better utilize the spatial relationship information in HSI. Similarly, Feng et al. [20] developed an algorithm for HAD based on local spatial constraint and TV regularization to fully explore the local structural information of the background. It should be emphasized that the aforementioned citations [18], [19], [20] all use k-means to construct a background dictionary. However, this background dictionary construction method has two disadvantages. A key parameter k in the k-means algorithm cannot be determined accurately. In addition, the clustering results are random, causing the constructed dictionary to be unstable. To solve this problem, some other dictionary construction methods are proposed, such as endmember dictionary construction methods [21], [47] and background and abnormal double dictionary construction methods [27], [28], [29], [30], [48].

B. Tensor-Based Representation Methods

Depending on whether a background dictionary needs to be constructed, tensor-based representation approaches can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

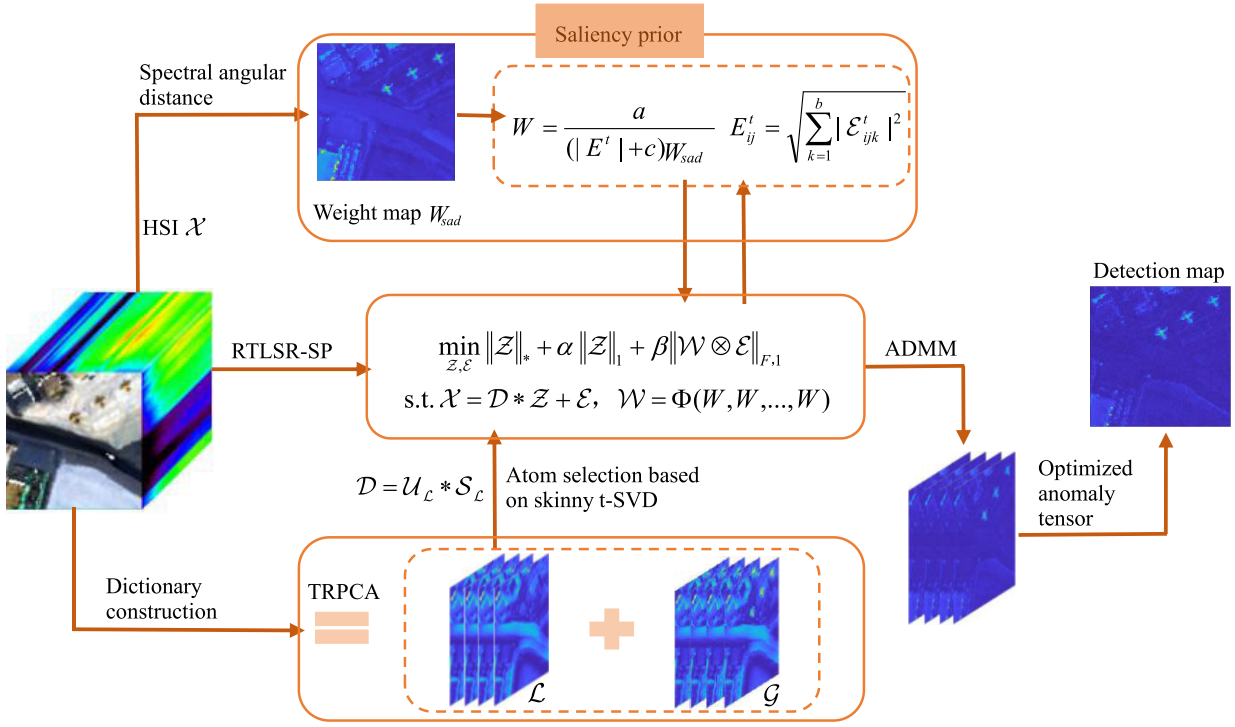


Fig. 1. Flowchart of the proposed RTLSR-SP approach for HAD.

subdivided into tensor decomposition-based approaches [14], [37], [38], [39] and TLRR-based approaches [32], [33], [34]. For example, Li et al. [14] provided a priori-based tensor approximation (PTA) method for HAD. However, PTA solely explores the low-rank properties of the background in the spectral dimension, disregarding its low-rank properties existing in the spatial dimension. In order to accurately estimate the low-rank background component, some sparsity norms for metric tensor data are developed. For instance, TNN is employed as a convex approximation of the nonconvex tensor tube rank to estimate the low-rank background component [15], [33]. Subsequently, some variant norms based on TNN including weighted TNN [31], ϵ-TNN [32], and weighted tensor schatten p-norm [37] are generated to fully explore the low-rank characteristics of the background. In order to explore the global low-rank and local piecewise smoothness of the background component, Sun et al. [34] proposed a tensor average rank and piecewise smoothing constraint (LARTVAD) method for HAD. However, LARTVAD does not take into account the inherent piecewise smoothness of the background in the spectral dimension. To tackle this issue, Feng et al. [40] employ a combination of tensor ring decomposition and factor TV regularization to exploit the low-rank and piecewise smoothness properties of the background component that is present in both spatial and spectral dimensions. Although these detectors maintain the multiway structure of HSI tensor data, most of them impose various types of prior knowledge constraints on the background while neglecting the exploration of anomaly prior information, thereby diminishing the performance of anomaly detection.

III. Proposed Method

For a tensor $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, let $\overline{\boldsymbol{\mathcal{A}}}$ denote the discrete Fourier transformation (DFT) result of $\boldsymbol{\mathcal{A}}$ along the third dimension, which can be obtained through the MATLAB command fft, that is, $\overline{\boldsymbol{\mathcal{A}}} = \text{fft}(\boldsymbol{\mathcal{A}}, [], 3)$. Similarly, using the inverse DFT, we have $\boldsymbol{\mathcal{A}} = \text{ifft}(\overline{\boldsymbol{\mathcal{A}}}, [], 3)$. For convenience, some commonly used notations are summarized in Nomenclature.

Fig. 1 offers the flowchart of the proposed approach, which comprises three main components, namely, the proposal of the RTLSR-SP model, a new sparse saliency prior weight map design method, and a new robust dictionary construction method. Subsequently, we will introduce these three parts in detail and give the entire optimization process of the proposed method.

A. Problem Formulation

Since there are mixed pixels in HSIs, all pixels in hyperspectral data are actually located in the union of multiple linear subspaces. In this case, considering that the RPCA model only assumes a single low-rank linear subspace as the underlying structure of the observed data, the LRR model is more appropriate than the RPCA model for handling real HSI data. The objective function of the LRR model can be expressed as follows:

$$\begin{aligned} &\min_{\boldsymbol{Z},\boldsymbol{E}} \ \|\boldsymbol{Z}\|_* + \alpha\|\boldsymbol{E}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{D}\boldsymbol{Z} + \boldsymbol{E} \end{aligned} \tag{1}$$

where $\boldsymbol{D}\boldsymbol{Z}$ denotes the background component and $\boldsymbol{E}$ represents the abnormal component.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

Given 3-D HSI data $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, where h and w represent the size of the spatial dimension and b represents the number of bands in the spectral dimension, the TLRR model is defined as follows:

$$\begin{aligned} &\min \ \|\mathcal{Z}\|_\star + \lambda\|\mathcal{E}\|_\ell \\ &\text{s.t. } \mathcal{X} = \mathcal{D} * \mathcal{Z} + \mathcal{E} \end{aligned} \tag{2}$$

where $\mathcal{D}$ is the background dictionary tensor that spans the entire data space and $\mathcal{Z}$ is the coefficient tensor of $\mathcal{D}$, also referred to as the LRR of $\mathcal{X}$ with respect to $\mathcal{D}$. The TNN of $\mathcal{Z}$ is defined as the sum of the singular values of all frontal slices of $\overline{\mathcal{Z}}$, denoted as $\|\mathcal{Z}\|_\star = (1/b)\sum_{i=1}^{b}\|\overline{\mathcal{Z}}^{(i)}\|_*$. The tensor ℓ norm is employed to represent the anomaly $\mathcal{E}$.

In [49], it was shown that low-rank constraints can effectively capture the global low-rank nature of the data. However, some detailed features of ground objects in HSI are often hidden in their local structures. In order to more accurately represent the background component in HSI tensor data, it is necessary to further mine the local structure information of the background component. Inspired by [18] and [42], we introduce a sparse regularization term in (2) as follows:

$$\begin{aligned} &\min \ \|\mathcal{Z}\|_\star + \alpha\|\mathcal{Z}\|_1 + \beta\|\mathcal{E}\|_\ell \\ &\text{s.t. } \mathcal{X} = \mathcal{D} * \mathcal{Z} + \mathcal{E}. \end{aligned} \tag{3}$$

Some existing TLRR-based HAD methods [31], [32], [33] focus on the mining of prior knowledge of background components, while the utilization of prior information of abnormal components is limited, which may lead to limited detection performance. In order to comprehensively exploit the spatial–spectral structural information in the HSI tensor data, we design a sparse saliency prior weight tensor for the anomaly tensor, aiming to suppress the background while enhancing the saliency of anomalous targets. In addition, tensor $\ell_{F,1}$-norm [50] is incorporated into our method to enhance the group sparsity of anomalous targets. The proposed RTLSR-SP model is given as follows:

$$\begin{aligned} &\min \ \|\mathcal{Z}\|_\star + \alpha\|\mathcal{Z}\|_1 + \beta\|\mathcal{W}\otimes\mathcal{E}\|_{F,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{D} * \mathcal{Z} + \mathcal{E}, \mathcal{W} = \Phi(\boldsymbol{W}, \boldsymbol{W}, \ldots, \boldsymbol{W}) \end{aligned} \tag{4}$$

where α and β are two regularization parameters, $\Phi(\cdot)$ means to stack multiple $\boldsymbol{W}$'s into a third-order tensor $\mathcal{W}$, and $\otimes$ represents the Hadamard product. The $\ell_{F,1}$-norm of tensor $\mathcal{E}$ is defined $\|\mathcal{E}\|_{F,1} = \sum_{i=1}^{h}\sum_{j=1}^{w}\|\mathcal{E}(i,j,:)\|_F$. The design of the sparse saliency prior weight map $\boldsymbol{W}$ and the construction of the robust background dictionary $\mathcal{D}$ will be presented in Sections III-B and III-C, respectively.

B. Sparse Saliency Prior Weight Map

In HAD, abnormal targets are regarded as salient targets in the visual system, and a saliency mechanism is introduced [25]. In this section, we first introduce two types of saliency prior weight maps, namely, an adaptive weight map and an iterative sparse reweighting weight map. Then, a sparse saliency prior weight map is designed to represent the saliency weights to distinguish anomalies from the background.

1) Adaptive Weight Map: In order to suppress the noise and background, an adaptive weight map is introduced. The adaptive weight map is obtained by applying the spectral angular distance [51] on the raw HSI tensor data $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, which effectively utilizes the spectral information contained in the HSI. The spectral angular distance is calculated as follows:

$$\mathrm{w}_g = \arccos\left[\frac{(\boldsymbol{\mathcal{X}}_g)^*\boldsymbol{\mu}}{\sqrt{(\boldsymbol{\mathcal{X}}_g)^*\boldsymbol{\mathcal{X}}_g\sqrt{\boldsymbol{\mu}^*\boldsymbol{\mu}}}}\right], \quad g = 1, 2, \ldots, h\times w \tag{5}$$

where $\boldsymbol{\mathcal{X}}_g$ is the gth pixel to be measured in the original HSI, $\boldsymbol{\mu}$ is the spectral mean of all pixels in the original HSI, and w_g is the weight value of the gth pixel. Then, an adaptive weight map can be obtained, denoted by $\boldsymbol{W}_{\mathrm{sad}}$.

To convert the angular distance into a similarity measure, we take the reciprocal of $\boldsymbol{W}_{\mathrm{sad}}$ to get $\boldsymbol{W}_{\mathrm{pr}}$ as follows:

$$\boldsymbol{W}_{\mathrm{pr}} = \frac{\mathbf{1}}{\boldsymbol{W}_{\mathrm{sad}}} \tag{6}$$

where $\mathbf{1}$ is a matrix whose elements are all 1. Equation (6) shows that, when $\boldsymbol{W}_{\mathrm{pr}}$ is smaller, the spectral difference between the two pixels is larger, which means that it is an abnormal target.

2) Iterative Sparse Reweighting Weight Map: Candes et al. [52] developed a reweighting ℓ_1-norm minimization method with weights inversely proportional to the true signal values. Inspired by this, we adopt an iterative reweighting strategy to enhance the sparsity of abnormal targets. Suppose that $\boldsymbol{E}^t$ is the anomaly matrix in the tth optimization. The iteratively sparse reweighting weight map is defined as follows:

$$\boldsymbol{W}_{\mathrm{sw}}^t = \frac{a}{|\boldsymbol{E}^t| + c} \tag{7}$$

where a is set to 2 in this article, c is a small positive value, and $\boldsymbol{E}_{ij}^t = (\sum_{k=1}^{b}|\boldsymbol{\mathcal{E}}_{ijk}^t|^2)^{1/2}$.

Based on (6) and (7), we define a sparse saliency prior weight map as follows:

$$\boldsymbol{W} = \boldsymbol{W}_{\mathrm{pr}} \otimes \boldsymbol{W}_{\mathrm{sw}}^t \tag{8}$$

where $\otimes$ represents the Hadamard product. Small weights retain nonzero elements, while large weights block nonzero elements, thereby better separating abnormal targets from the background.

C. Robust Dictionary Construction

In the TLRR-based HAD method, the quality of the background dictionary plays an irreplaceable role in improving detection performance. However, some existing algorithms treat the entire HSI tensor data as a background dictionary, such as [32], [33], and [35]. This type of dictionary usually has two disadvantages. On the one hand, HSI is inevitably contaminated by noise and outliers during the acquisition and transmission process. On the other hand, the dictionary contains a vast number of atoms, which leads to significant computational overhead. In order to improve the robustness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

of the dictionary to noise and outliers, we adopt the TRPCA algorithm as follows:

$$\min_{\mathcal{L},\mathcal{G}} \ \|\mathcal{L}\|_\star + \lambda\|\mathcal{G}\|_1 \quad \text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{G} \tag{9}$$

where λ is the regularization parameter to balance low-rank and sparse terms. $\mathcal{L}$ and $\mathcal{G}$ denote the low-rank background tensor and the sparse anomaly tensor, respectively. It is worth noting that, if we set the background dictionary tensor $\mathcal{D}$ in (4) to be $\mathcal{L}$, then we have $\mathcal{Z} \in \mathbb{R}^{w\times w\times b}$. When w is large, the computational cost of directly applying ADMM [53] to solve (4) is very high because each iteration requires the calculation of b SVD on the $w \times w$ matrices. To improve the computational efficiency of the RTLSR-SP approach, we construct the background dictionary tensor by calculating the skinny t-SVD of $\mathcal{L}$, i.e., $\mathcal{L} = \mathcal{U}_\mathcal{L} * \mathcal{S}_\mathcal{L} * \mathcal{V}_\mathcal{L}^*$. As a result, the background component in (4) can be transformed into $\mathcal{D}*\mathcal{Z} = \mathcal{U}_\mathcal{L}*\mathcal{S}_\mathcal{L}*\mathcal{V}_\mathcal{L}^**\mathcal{Z}'$. Then, we represent $\mathcal{U}_\mathcal{L}*\mathcal{S}_\mathcal{L}$ and $\mathcal{V}_\mathcal{L}^**\mathcal{Z}'$ as dictionary tensor $\mathcal{D} \in \mathbb{R}^{h\times r_\mathcal{L}\times b}$ and coefficient tensor $\mathcal{Z} \in \mathbb{R}^{r_\mathcal{L}\times w\times b}$, where the tensor tube rank of $\mathcal{L}$ satisfies $r_\mathcal{L} \ll w$. Consequently, in each iteration, the SVD calculation amount of the matrix will be reduced from $w \times w$ to $r_\mathcal{L} \times w$. For a detailed proof of the equivalence of the minimization pair $(\mathcal{Z}, \mathcal{E})$ and the minimization pair $(\mathcal{Z}', \mathcal{E})$, please refer to the literature [44]. The detailed process of robust dictionary construction is summarized in Algorithm 1.

In addition, an adaptive method to estimate tensor tube rank $r_\mathcal{L}$ is designed. We first obtain $\overline{\mathcal{L}}$ through DFT and then calculate the nonzero singular values $\{\sigma_1^i, \ldots, \sigma_{c_i}^i\}$ of each front slice $\overline{\boldsymbol{L}}^{(i)}$ of $\overline{\mathcal{L}}$ in descending order. Then, let $\boldsymbol{a}_k^i = \sigma_k^i/\sigma_{k+1}^i (1 \le k \le c_i - 1)$, $k_*^i = \arg\max_k \boldsymbol{a}_k^i$, and $\boldsymbol{a}_*^i = \boldsymbol{a}_{k_*^i}^i$. If $(c_i - 1)\boldsymbol{a}_*^i / \sum_{k\neq k_*^i} \boldsymbol{a}_k^i < 10$ (need not truncation), then let $k_*^i = c_i$. Finally, we get the estimated tensor tube rank $r_\mathcal{L} = \max(k_*^1, \ldots, k_*^b)$ of tensor $\mathcal{L}$.

Algorithm 1 Pseudocode for Robust Dictionary Construction

Input: HSI data $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, parameter λ.
1: **Initialization:** $t = 0$, $\mathcal{L}^0 = \mathcal{G}^0 = \mathcal{Q}^0 = 0$, $\rho = 1.2$, $\tau^0 = 1e\text{-}4$, $\tau_{\max} = 1e10$, $\epsilon_1 = 1e\text{-}3$.
2: **While** not converged **do**;
3: Update $\mathcal{L}^{t+1}$ by
$\mathcal{L}^{t+1} = \arg\min_{\mathcal{L}} \|\mathcal{L}\|_\star + \frac{\tau^t}{2}\|\mathcal{X} - \mathcal{L} - \mathcal{G}^t + \frac{\mathcal{Q}^t}{\mu^t}\|_F^2$;
4: Update $\mathcal{G}^{t+1}$ by
$\mathcal{G}^{t+1} = \arg\min_{\mathcal{G}} \lambda\|\mathcal{G}\|_1 + \frac{\tau^t}{2}\|\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{G} + \frac{\mathcal{Q}^t}{\mu^t}\|_F^2$;
5: Update the Lagrangian multiplier $\mathcal{Q}$ and penalty parameter τ by
$\mathcal{Q}^{t+1} = \mathcal{Q}^t + \tau^t(\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{G}^{t+1})$;
$\tau^{t+1} = \min(\rho\tau^t, \tau_{\max})$;
6: The convergence condition is checked by
Error $= \max(\|\mathcal{L}^{t+1} - \mathcal{L}^t\|_\infty \le \epsilon_1, \|\mathcal{G}^{t+1} - \mathcal{G}^t\|_\infty \le \epsilon_1, \|\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{G}^{t+1}\|_\infty \le \epsilon_1)$;
7: Update iteration: $t = t + 1$;
8: **End while**
9: Obtain a low-rank clean tensor $\mathcal{L}$.
10: Construct the dictionary $\mathcal{D} = \mathcal{U}_\mathcal{L} * \mathcal{S}_\mathcal{L}$ with the help of the skinny t-SVD of $\mathcal{L}$, i.e., $\mathcal{L} = \mathcal{U}_\mathcal{L} * \mathcal{S}_\mathcal{L} * \mathcal{V}_\mathcal{L}^*$;
Output: Robust dictionary tensor $\mathcal{D} \in \mathbb{R}^{h\times r_\mathcal{L}\times b}$.

D. Optimizing the RTLSR-SP Model

The ADMM-based optimization algorithm will be employed in this section to solve problem (4). By introducing two auxiliary variables $\mathcal{B}$ and $\mathcal{C}$, problem (4) is rewritten as the following equivalent minimization problem:

$$\begin{aligned}&\min \ \|\mathcal{B}\|_\star + \alpha\|\mathcal{C}\|_1 + \beta\|\mathcal{W}\otimes\mathcal{E}\|_{F,1}\\ &\text{s.t. } \mathcal{X} = \mathcal{D}*\mathcal{Z} + \mathcal{E}, \mathcal{W} = \Phi(W, W, \ldots, W)\\ &\quad \mathcal{B} = \mathcal{Z}, \mathcal{C} = \mathcal{Z}.\end{aligned} \tag{10}$$

Then, the augmented Lagrange function corresponding to problem (10) is offered as follows:

$$\begin{aligned}&\Psi(\mathcal{Z}, \mathcal{B}, \mathcal{C}, \mathcal{E}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mu)\\ &= \|\mathcal{B}\|_\star + \alpha\|\mathcal{C}\|_1\\ &\quad + \beta\|\mathcal{W}\otimes\mathcal{E}\|_{F,1} + \langle\mathcal{Y}_1, \mathcal{X} - \mathcal{D}*\mathcal{Z} - \mathcal{E}\rangle + \langle\mathcal{Y}_2, \mathcal{B} - \mathcal{Z}\rangle\\ &\quad + \langle\mathcal{Y}_3, \mathcal{C} - \mathcal{Z}\rangle + \frac{\mu}{2}\|\mathcal{X} - \mathcal{D}*\mathcal{Z} - \mathcal{E}\|_F^2 + \frac{\mu}{2}\|\mathcal{B} - \mathcal{Z}\|_F^2\\ &\quad + \frac{\mu}{2}\|\mathcal{C} - \mathcal{Z}\|_F^2\end{aligned} \tag{11}$$

where μ is the penalty parameter and $\mathcal{Y}_i (i = 1, 2, 3)$ are the Lagrangian multipliers. Each variable in the augmented Lagrange function (11) can be optimized alternately, while the other variables are fixed.

After several simple derivations, the following equivalent problems can be obtained:

$$\begin{aligned}&\Psi(\mathcal{Z}, \mathcal{B}, \mathcal{C}, \mathcal{E}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mu)\\ &= \|\mathcal{B}\|_\star + \alpha\|\mathcal{C}\|_1\\ &\quad + \beta\|\mathcal{W}\otimes\mathcal{E}\|_{F,1} + \frac{\mu}{2}\left\|\mathcal{X} - \mathcal{D}*\mathcal{Z} - \mathcal{E} + \frac{\mathcal{Y}_1}{\mu}\right\|_F^2\\ &\quad + \frac{\mu}{2}\left\|\mathcal{B} - \mathcal{Z} + \frac{\mathcal{Y}_2}{\mu}\right\|_F^2 + \frac{\mu}{2}\left\|\mathcal{C} - \mathcal{Z} + \frac{\mathcal{Y}_3}{\mu}\right\|_F^2\end{aligned} \tag{12}$$

1) *Updating of* $\mathcal{Z}$ By fixing other variables, the following problem needs to be minimized:

$$\begin{aligned}\mathcal{Z}^{t+1} &= \arg\min_{\mathcal{Z}} \frac{\mu^t}{2}\left\|\mathcal{X} - \mathcal{D}*\mathcal{Z} - \mathcal{E}^t + \frac{\mathcal{Y}_1^t}{\mu^t}\right\|_F^2\\ &\quad + \frac{\mu^t}{2}\left\|\mathcal{B}^t - \mathcal{Z} + \frac{\mathcal{Y}_2^t}{\mu^t}\right\|_F^2 + \frac{\mu^t}{2}\|\mathcal{C}^t - \mathcal{Z} + \frac{\mathcal{Y}_3^t}{\mu^t}\|_F^2.\\ &= \arg\min_{\mathcal{Z}} \left\|\mathcal{D}*\mathcal{Z} - \mathcal{P}_1^t\right\|_F^2 + \left\|\mathcal{Z} - \mathcal{P}_2^t\right\|_F^2\\ &\quad + \left\|\mathcal{Z} - \mathcal{P}_3^t\right\|_F^2\end{aligned} \tag{13}$$

where $\mathcal{P}_1^t = \mathcal{X} - \mathcal{E}^t + \mathcal{Y}_1^t/\mu^t$, $\mathcal{P}_2^t = \mathcal{B}^t + \mathcal{Y}_2^t/\mu^t$, $\mathcal{P}_3^t = \mathcal{C}^t + \mathcal{Y}_3^t/\mu^t$, and t represents the number of iterations. Problem (13) can be realized by the following linear system:

$$\mathcal{Z}^{t+1} = \left(\mathcal{D}^**\mathcal{D} + 2\mathcal{I}\right)^{-1} * \left(\mathcal{D}^**\mathcal{P}_1^t + \mathcal{P}_2^t + \mathcal{P}_3^t\right). \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

2) *Updating of* $\mathcal{B}$: By fixing other variables, we need to optimize the following problem to update $\mathcal{B}$:

$$\begin{aligned}\mathcal{B}^{t+1} &= \arg\min_{\mathcal{B}} \|\mathcal{B}\|_\star + \frac{\mu^t}{2}\|\mathcal{B} - \mathcal{Z}^{t+1} + \frac{\mathcal{Y}_2^t}{\mu^t}\|_F^2 \\ &= \arg\min_{\mathcal{B}} \|\mathcal{B}\|_\star + \frac{\mu^t}{2}\left\|\mathcal{B} - \mathcal{P}_4^t\right\|_F^2 \end{aligned} \tag{15}$$

where $\mathcal{P}_4^t = \mathcal{Z}^{t+1} - (\mathcal{Y}_2^t/\mu^t)$; by calculating $\mathcal{P}_4^t = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$, a closed-form solution to problem (15) can be obtained as follows:

$$\mathcal{B}^{t+1} = \mathcal{U} * \mathcal{S}_{\frac{1}{\mu^t}} * \mathcal{V}^* \tag{16}$$

where

$$\begin{cases} \mathcal{S}_{\frac{1}{\mu^t}} = \text{ifft}\left(\overline{\mathcal{S}}_{\frac{1}{\mu^t}}, [], 3\right) \\ \overline{\mathcal{S}}_{\frac{1}{\mu^t}}(i,j,k) = \max\left(\overline{\mathcal{S}}(i,j,k) - \dfrac{1}{\mu^t}, 0\right) \\ \overline{\mathcal{S}} = \text{fft}(\mathcal{S}, [], 3). \end{cases} \tag{17}$$

3) *Updating of* $\mathcal{C}$: In (12), delete other items that are not related to $\mathcal{C}$, and then, the following minimization problem can be obtained:

$$\begin{aligned}\mathcal{C}^{t+1} &= \arg\min_{\mathcal{C}} \alpha\|\mathcal{C}\|_1 + \frac{\mu^t}{2}\left\|\mathcal{C} - \mathcal{Z}^{t+1} + \frac{\mathcal{Y}_3^t}{\mu^t}\right\|_F^2 \\ &= \arg\min_{\mathcal{C}} \alpha\|\mathcal{C}\|_1 + \frac{\mu^t}{2}\left\|\mathcal{C} - \mathcal{P}_5^t\right\|_F^2 \end{aligned} \tag{18}$$

where $\mathcal{P}_5^t = \mathcal{Z}^{t+1} - \mathcal{Y}_3^t/\mu^t$.
Problem (18) can be solved by the elementwise shrinkage operator $\Theta_o^{(r)}$ to obtain a closed-form solution as follows:

$$\mathcal{C}^{t+1} = \Theta_{\alpha/\mu^t}\left(\mathcal{P}_5^t\right) \tag{19}$$

where $\Theta_o^{(r)} = \text{sign}(r) \cdot \max(\text{abs}(r) - o, 0)$.

Algorithm 2 Pseudocode of RTLSR-SP Model

Input: HSI data $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, background dictionary tensor $\mathcal{D}$, parameters α and β.
1: **Initialization:** $t = 0, \mathcal{E}^0 = \mathcal{Y}_1^0 = 0, \mathcal{B}^0 = \mathcal{C}^0 = \mathcal{Y}_2^0 = \mathcal{Y}_3^0 = 0, \mu^0 = 1e\text{-}1, \epsilon_2 = 1e\text{-}3, \rho = 1.2, \mu_{\max} = 1e8$, maxiter $= 100$;
2: **While** $0 \le t \le$ maxiter or not converged **do**;
3: Update $\mathcal{Z}^{t+1}$ by (14);
4: Update $\mathcal{B}^{t+1}$ by (16);
5: Update $\mathcal{C}^{t+1}$ by (19);
6: Update $\mathcal{E}^{t+1}$ by (21);
7: Update $\mathcal{W}^{t+1}$ by (22);
8: Update the Lagrangian multipliers $\mathcal{Y}_i (i = 1, 2, 3)$ and penalty parameter μ by (24).
9: The convergence condition is checked by (25).
10: Update iteration: $t = t + 1$;
End while
11: Obtain the optimal abnormal tensor $\mathcal{E}_*$;
12: Calculate the anomaly detection map $\boldsymbol{T}$ by $\boldsymbol{T}_{ij} = \sqrt{\sum_{k=1}^{b} |\mathcal{E}_*(i,j,k)|^2}$;
Output: Anomaly detection map $\boldsymbol{T}$.

4) *Updating of* $\mathcal{E}$: Extracting all terms related to $\mathcal{E}$ from (12), the following optimization problem can be realized:

$$\mathcal{E}^{t+1} = \arg\min_{\mathcal{E}} \beta\|\mathcal{W}^t \otimes \mathcal{E}\|_{F,1} + \frac{\mu^t}{2}\|\mathcal{E} - \mathcal{P}_6^t\|_F^2 \tag{20}$$

where $\mathcal{P}_6^t = \mathcal{X} - \mathcal{D} * \mathcal{Z}^{t+1} + (\mathcal{Y}_1^t/\mu^t)$.
Problem (20) can get a closed-form solution through [49], and the kth band of $\mathcal{E}$ is given as follows:

$$\mathcal{E}(:,:,k)^{t+1} = \begin{cases} \dfrac{\|\mathcal{P}_6^t(:,:,k)\|_F - \frac{W^t\beta}{\mu^t}}{\|\mathcal{P}_6^t(:,:,k)\|_F}\mathcal{P}_6^t(:,:,k), & \dfrac{W^t\beta}{\mu^t} < \|\mathcal{P}_6^t(:,:,k)\|_F \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

5) *Updating of* $\mathcal{W}$: The sparse saliency prior weight tensor $\mathcal{W}$ is updated as follows:

$$\mathcal{W}^{t+1} = \Phi\left(\boldsymbol{W}^{t+1}, \boldsymbol{W}^{t+1}, \ldots, \boldsymbol{W}^{t+1}\right) \tag{22}$$

where

$$\boldsymbol{W}^{t+1} = \boldsymbol{W}_{\text{pr}} \otimes \boldsymbol{W}_{\text{sw}}^t. \tag{23}$$

6) Update Lagrange multipliers $\mathcal{Y}_i (i = 1, 2, 3)$ and penalty parameter μ

$$\begin{cases} \mathcal{Y}_1^{t+1} = \mathcal{Y}_1^t + \mu^t\left(\mathcal{X} - \mathcal{D} * \mathcal{Z}^{t+1} - \mathcal{E}^{t+1}\right) \\ \mathcal{Y}_2^{t+1} = \mathcal{Y}_2^t + \mu^t\left(\mathcal{B}^{t+1} - \mathcal{Z}^{t+1}\right) \\ \mathcal{Y}_3^{t+1} = \mathcal{Y}_3^t + \mu^t\left(\mathcal{C}^{t+1} - \mathcal{Z}^{t+1}\right) \\ \mu^{t+1} = \min\left(\rho\mu^t, \mu_{\max}\right). \end{cases} \tag{24}$$

7) The convergence condition is defined as follows:

$$\text{Error} = \max\left\{\begin{matrix} \left\|\mathcal{Z}^{t+1} - \mathcal{Z}^t\right\|_\infty, \left\|\mathcal{B}^{t+1} - \mathcal{B}^t\right\|_\infty \\ \left\|\mathcal{C}^{t+1} - \mathcal{C}^t\right\|_\infty, \left\|\mathcal{E}^{t+1} - \mathcal{E}^t\right\|_\infty \\ \left\|\mathcal{X} - \mathcal{D} * \mathcal{Z}^{t+1} - \mathcal{E}^{t+1}\right\|_\infty \end{matrix}\right\} \le \epsilon_2. \tag{25}$$

In Algorithm 2, the whole process of ADMM solving problem (10) is summarized.

IV. Experiments and Analysis of Results

In this section, extensive experiments are conducted on one simulated HSI dataset and six real HSI datasets to verify the effectiveness and superiority of the proposed RTLSR-SP algorithm. Before feeding the data into Algorithms 1 and 2, we perform band-by-band normalization on the raw data. All numerical experiments are implemented in MATLAB 2022a on the computer with Intel[1] Core[2] i7-12700F processors, 2.10-GHz CPU, and 32 GB of memory.

A. Experimental Datasets

1) Salinas Dataset: This simulated dataset is provided by Tan et al. [54], which is generated based on the Salinas

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

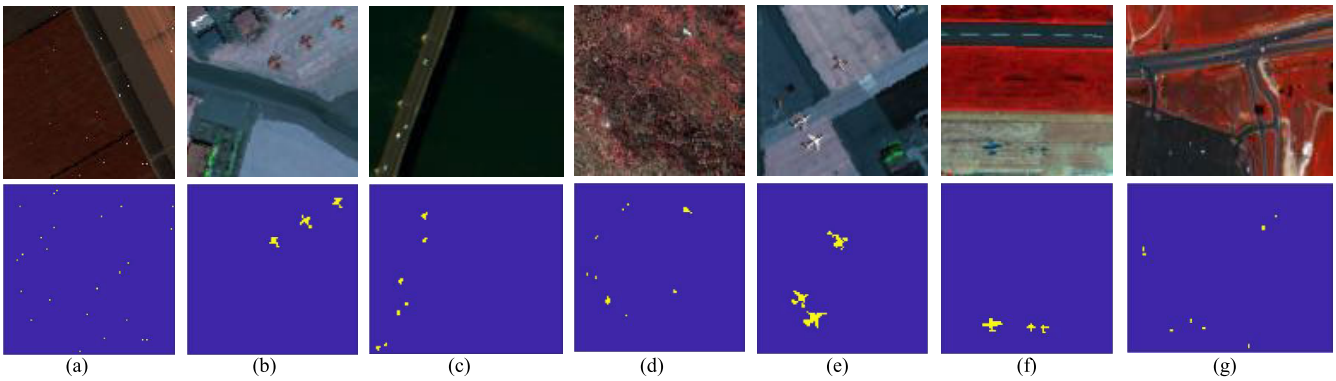


Fig. 2. Pseudocolor images and ground-truth maps of seven hyperspectral datasets. (a) Salinas. (b) San Diego-I. (c) Pavia Center. (d) Hyperion. (e) San Diego-II. (f) Gulfport. (g) HYDICE.

TABLE I
DESCRIPTION OF SEVEN DATASETS

Datasets	Spatial size	Location	Spatial resolution	Number of bands	Sensors	Image size	Size of targets	Complexity of background
Salinas	120 × 120	Salinas Valley	3.7	204	AVIRIS	small	small	complex
San Diego-I	100 × 100	San Diego	3.5	189	AVIRIS	small	large	complex
Pavia Centor	105 × 120	Italy	1.3	102	ROSIS	small	small	simple
Hyperion	100 × 100	Okavango Delta	30	145	Hyperion	small	small	complex
San Diego-II	100 × 100	San Diego	3.5	189	AVIRIS-II	small	large	complex
Gulfport	100 × 100	Gulfport	3.4	191	AVIRIS	small	large and small	complex
HYDICE	80 × 121	Urban	1.5	162	HYDICE	small	small	complex

scene, acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor from Salinas Valley, CA, USA.[3] Fig. 2(a) displays the pseudocolor image and the corresponding ground-truth map of the Salinas dataset.

2) San Diego-I Dataset: The dataset was obtained by the AVIRIS sensor at San Diego airport, USA. The abnormal targets are three planes with a total of 134 anomalous pixels. The pseudocolor map and the corresponding ground-truth map of the San Diego-I dataset are shown in Fig. 2(b).

3) Pavia Center Dataset: The dataset was collected by the reflective optics system imaging spectrometer (ROSIS) at the Pavia Center, Italy. The vehicles on the bridge and the exposed soil near the piers are regarded as anomalous targets. The pseudocolor map and the corresponding ground-truth map of the Pavia Center dataset are stated in Fig. 2(c).

4) Hyperion Dataset: The dataset was obtained from above the Okavango Delta by the EO-I Hyperion sensor. The background mainly consists of observation data of 14 different land cover types, with a total of 32 anomalous pixels. The pseudocolor map and the corresponding ground-truth map of the Hyperion dataset are provided in Fig. 2(d).

5) San Diego-II Dataset: The dataset was captured by the AVIRIS-II sensor from the area over San Diego, CA, USA. The image scene consists of grass, an airstrip, and three aircrafts that are considered anomalous targets. The pseudocolor map and the corresponding ground-truth map of the San Diego-II dataset are displayed in Fig. 2(e).

6) Gulfport Dataset: This dataset was taken by the AVIRIS sensor in the Gulfport area, Southern Mississippi, USA, in 2010. The image scene mainly consists of airport runways, highways, vegetation, and three aircraft marked as abnormal targets. Fig. 2(f) displays the pseudocolor map and the corresponding ground-truth map of the Gulfport dataset.

7) HYDICE Dataset: The dataset was shot by the hyperspectral digital imagery collection experiment (HYDICE) sensor over an urban area in CA, USA. The background of the image mainly consists of a parking lot, water, land, and two roads. A total of 21 pixels containing cars and roofs are defined as abnormal targets. The pseudocolor map and the corresponding ground-truth map of the HYDICE dataset are depicted in Fig. 2(g).

A brief summary of these datasets is provided in Table I.

B. Comparison Algorithms

Eight state-of-the-art comparison algorithms are selected in the experiments to demonstrate the superiority of the proposed RTLSR-SP algorithm. These algorithms include two statistics-based methods [i.e., the classic LRX [7] and an improved RX-based algorithm RX-segmented [8] (call SegRX for short)], one CR method (CRD) [24], two low-rank and SR methods (i.e., LRASR [18] and LSMAD [11]), and three tensor representation methods (i.e., PTA [14], LARTVAD [34], and PCA-TLRSR [31]).

C. Experimental Settings

For the eight comparison algorithms, we adopt the public source codes provided by the authors and fine-tune them according to the range of parameter settings in the corresponding literature to record the best performance. As shown in Table II, the specific parameter values of all algorithms on each dataset are given. The LRX algorithm incorporates

[3]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

two parameters, namely, the inner window size ω_{in} and the outer window size ω_{out}. The CRD algorithm has three parameters: inner window size ω_{in}, outer window size ω_{out}, and regularization parameter λ. The LRASR algorithm has four parameters: the number of clusters K, the number of pixels chosen in each cluster P, and the compromise parameters λ and β. The LSMAD algorithm involves a low-rank matrix of maximum rank r and a sparse matrix of cardinality k. The PTA algorithm involves five parameters, including truncated low-rank r, regularization parameters α and β, and nonnegative penalty parameters μ and τ. The LARTVAD algorithm contains three parameters: the number K of local clusters and the regularization parameters λ and β. The PCA-TLRSR algorithm contains four parameters: the dimension d of PCA preprocessing, the parameter λ_1 of constructing the dictionary, the weight value ω, and the regularization parameter λ_2. The parameter settings of the proposed RTLSR-SP algorithm and the impact of different parameters on the detection results are discussed in detail in Section IV-F.

D. Evaluation Metrics

The anomaly detection performance of each algorithm is assessed through both qualitative and quantitative evaluations in our experiments. Specifically, qualitative evaluation indicators include the anomaly detection map, the 3-D receiver operating characteristic (ROC) curve [55], the corresponding 2-D ROC curve, and the anomaly background separation map. The anomaly detection map can visually display the detection effect. The 3-D ROC curve is a function of the detection probability P_D, false alarm probability P_F, and decision threshold τ to accurately represent the comprehensive detection capability and background suppression ability of each anomaly detection algorithm. By decomposing the 3-D ROC curve, three corresponding 2-D ROC curves can be derived: (P_D, P_F), (P_D, τ), and (P_F, τ). The anomaly background separation map plots the pixel values of both the background and anomalous components in a detection result map within a designated box in a coordinate system. The larger the distance between the background box and the anomaly box, the better the separation performance of the background component and the anomaly component. The shorter the background box, the better the background suppression. A short background box and a large separation distance mean that the performance of the detector is optimal.

The quantitative evaluation index is the area under the ROC curve (AUC), where $\text{AUC}_{(D,F)}$ is a commonly used evaluation index to evaluate the effectiveness of the detector [17], [18], [19]. However, $\text{AUC}_{(D,F)}$ only considers the ROC relationship analysis between P_D and P_F and ignores the impact of the decision threshold τ, thus failing to reflect the comprehensive detection capability of the detector. In order to address this issue, Chang [55] developed various AUC evaluation metrics based on the 3-D ROC curve to comprehensively assess the detection performance of the detector. These AUC evaluation metrics include $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, $\text{AUC}_{(F,\tau)}$, AUC_{ODP}, AUC_{SNPR}, and AUC_{TDBS}, respectively, where AUC_{ODP}, AUC_{SNPR}, and AUC_{TDBS} are defined as follows:

$$\begin{cases} \text{AUC}_{\text{ODP}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \\ \text{AUC}_{\text{SNPR}} = \dfrac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}} \\ \text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)}. \end{cases} \tag{26}$$

The evaluation indicators $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$, respectively, represent the effectiveness of the detector, the capability of target detection, and the capability of background suppression. AUC_{ODP} evaluates the overall detection probability of the detector, AUC_{SNPR} evaluates the signal-to-noise probability ratio of the detector, and AUC_{TDBS} evaluates the combined capabilities of target detection and background suppression for the detector. In general, the smaller the value of $\text{AUC}_{(F,\tau)}$, the larger the other values, and the better the detection performance of the detector.

E. Experimental Results

1) Qualitative Comparison: From Fig. 3(a), it can be seen that, on the Salinas dataset, the LRX, SegRX, and PTA fail to detect the anomaly pixels. Compared with other competitors, CRD and LSMAD achieve better background suppression results, but some abnormal pixels are also suppressed, especially LSMAD. The PCA-TLRSR can detect a part of anomaly pixels, but it is subject to serious background interference. Compared with other detectors, LRASR, LARTVAD, and the proposed RTLSR-SP have a higher response to anomaly pixels. Furthermore, the RTLSR-SP outperforms LRASR and LARTVAD in terms of background suppression. For the San Diego-I dataset shown in Fig. 3(b), SegRX is unable to successfully detect three anomalous aircraft. The LRX, CRD, LSMAD, and LARTVAD have low response to anomalous aircraft. The LRASR, PTA, and PCA-TLRSR are able to detect all anomalous aircrafts but suffer from severe background pollution, especially PTA and PCA-TLRSR. In contrast, the proposed RTLSR-SP can well suppress the background while highlighting abnormal targets. From Fig. 3(c), we can easily observe that, on the Pavia Center dataset, except for LRX and PTA, other detectors can detect abnormal pixels. The SegRX and CRD only detect some abnormal pixels. Compared with LRASR, LARTVAD, and PCA-TLRSR, LSMAD achieved a better background suppression effect. Compared with all other detectors, the proposed RTLSR-SP is able to detect anomalous pixels well while suppressing the background. As can be seen in Fig. 3(d), on the Hyperion dataset, PTA fails to detect small anomalous targets. The abnormal responses seem to be similar in the detection maps of LRX, SegRX, CRD, and LRASR, but LRASR suppresses the background slightly better. The LSMAD can suppress the background well but also suppress some abnormal pixels, resulting in a low detection rate. The LARTVAD and PCA-TLRSR produce a large number of false alarms, which inevitably pollute the anomaly detection maps. Compared with other algorithms, the proposed RTLSR-SP algorithm can identify abnormal targets more clearly and suppress the background better.

For the San Diego-II dataset exhibited in Fig. 3(e), it can be found that all algorithms can successfully detect three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AUC VALUES OF VARIOUS ANOMALY DETECTORS USING DIFFERENT DATASETS

Methods	Parameters	Salinas	San Diego-I	Pavia Center	Hyperion	San Diego-II	Gulfport	HYDICE
LRX	w_{in}, w_{out}	(3,5)	(5,7)	(3,5)	(3,5)	(7,9)	(3,5)	(3,5)
CRD	w_{in}, w_{out}, λ	(5,7),1e-6	(17,19),1e-6	(3,5),1e-6	(5,7),1e-6	(15,17),1e-6	(3,9),1e-6	(13,15),1e-3
LRASR	K, P, λ, β	15,20,1,1	15,20,0.1,0.1	15,20,0.1,0.1	15,20,1,1	15,20,0.1,0.1	15,20,0.1,0.1	15,20,1,1
LSMAD	r, k	11,1e4	12,1e5	31,1e5	11,1e4	12,1e5	23,1e5	11,1e4
PTA	r, α, β	1,1e2,1e-1	1,1,1e-1	1,1,1e-1	1,1e4,1e-2	1,1e-4,1	1,1e-4,1	1,1e4,1e3
	μ, τ	1e-2,1	1e-4,1	1e-4,1e-1	1e4,1	1e-4,1	1e-4,1	1e4,1
LARTVAD	K, λ, β	120,1,3	80,2,10	80,0.01,10	120,0.05,5	120,2,5	80,2,10	120,10,1
PCA-TLRSR	$d, \lambda_1, \lambda_2, \omega$	15,0.06,1e-2,5	6,0.02,1e-2,5	15,0.06,1e-2,5	15,0.06,1e-2,5	6,0.02,1e-2,5	17,0.06,0.05,1	15,0.06,1e-2,5
RTLSR-SP	λ	0.001	0.007	0.001	0.003	0.01	0.007	0.01
	α, β	1e5,1e-5	1e-4,1e1	1e2,1e3	1e-1,1e-4	1e-2,1e0	1e-1,1e1	1e1,1e2

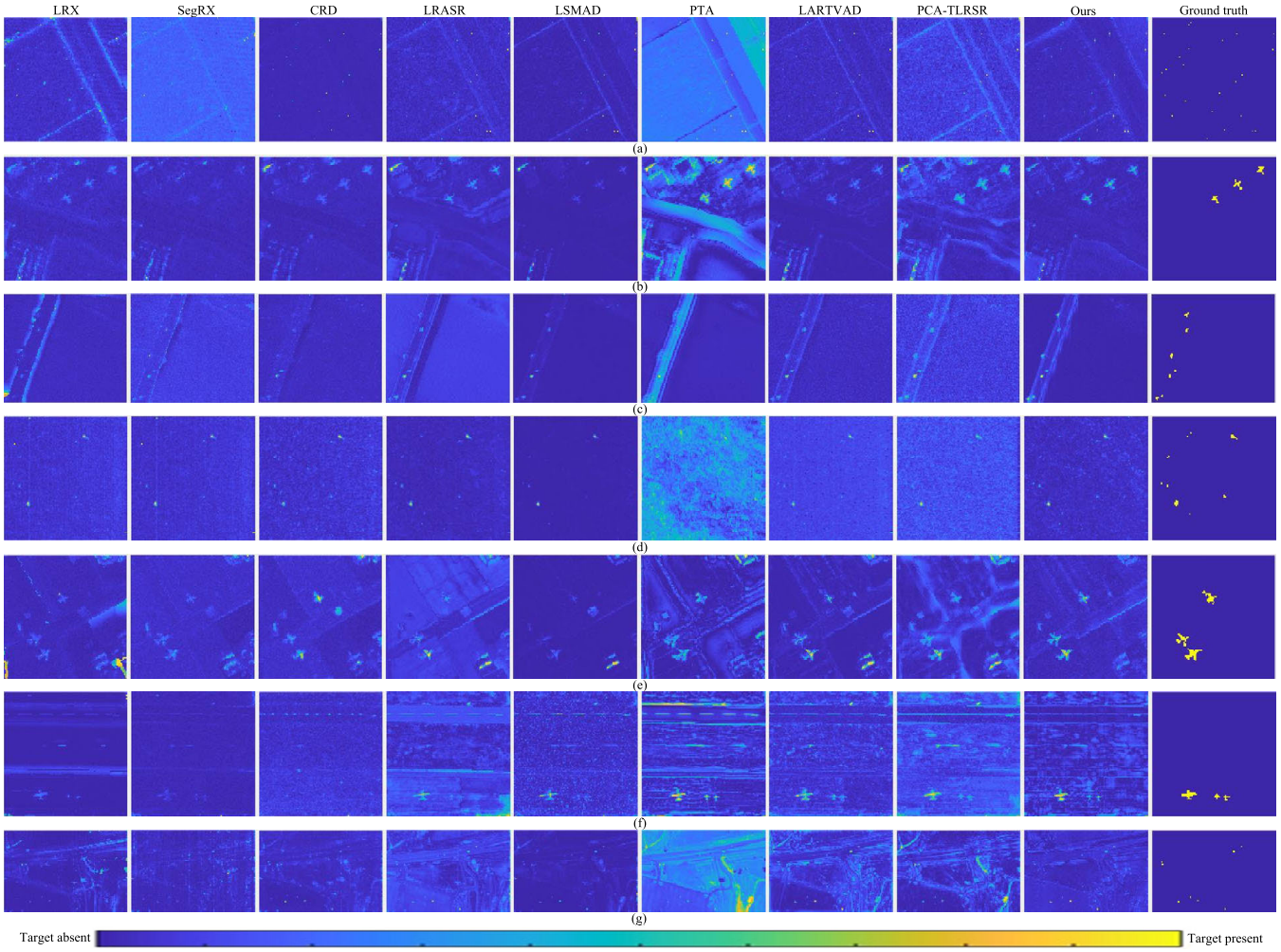


Fig. 3. Anomaly detection maps obtained by different anomaly detection algorithms and corresponding ground-truth maps on seven HSI datasets. (a) Salinas. (b) San Diego-I. (c) Pavia Center. (d) Hyperion. (e) San Diego-II. (f) Gulfport. (g) HYDICE.

abnormal aircraft. However, the characteristic shapes of abnormal aircraft in the detection maps of LRX, SegRX, LRASR, LSMAD, and LARTVAD are not all complete. The CRD, PTA, and PCA-TLRSR can detect all abnormal aircraft but cannot filter to the background. Compared with other algorithms, the proposed RTLSR-SP can highlight the three abnormal planes well and has strong background suppression ability. For the Gulfport dataset displayed in Fig. 3(f), the LRX, SegRX, LRASR, and LSMAD cannot successfully detect two small anomalous planes. The CRD fails to detect anomalous aircraft. The four tensor representation-based methods, PTA, LARTVAD, PCA-TLRSR, and the proposed RTLSR-SP, can all detect the three anomalous planes, among which the background suppression effect of RTLSR-SP is the best. For the HYDICE dataset displayed in Fig. 3(g), the PTA does not successfully detect anomalous targets. Compared with other detectors, LRASR, LARTVAD, and PCA-TLRSR are all heavily contaminated by background, resulting in poor detection results. LRX and SegRX can only detect some abnormal pixels. On the contrary, CRD, LSMAD, and RTLSR-SP can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

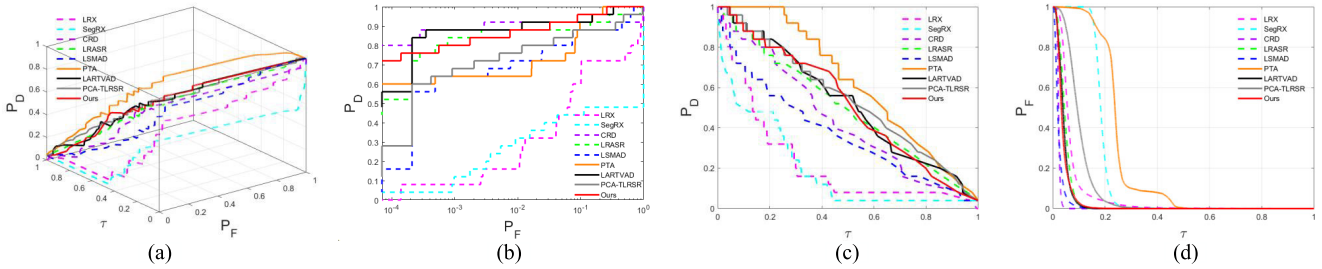


Fig. 4. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the Salinas dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

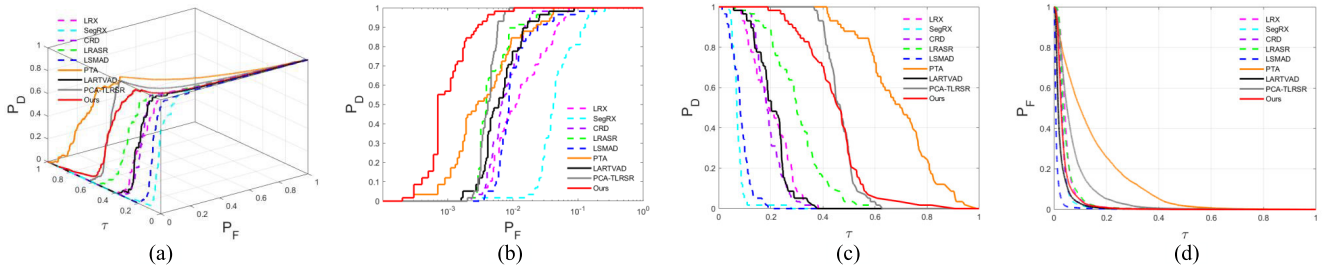


Fig. 5. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the San Diego-I dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

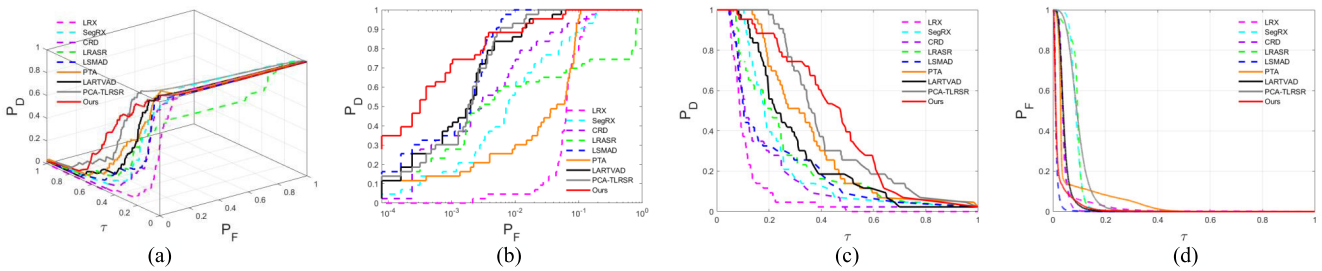


Fig. 6. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the Pavia Center dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

clearly detect abnormal pixels, but RTLSR-SP has a higher response to abnormal pixels.

The 2-D ROC curve of (P_D, P_F) exhibits better overall detection performance when it is closer to the upper left corner. As shown in Figs. 4(b)–10(b), from the 2-D ROC curves displayed on the San Diego-I, Pavia Center, San Diego-II, Gulfport, and HYDICE datasets, we can observe that the ROC curve of the RTLSR-SP algorithm is almost always higher than that of other algorithms, which indicates that the RTLSR-SP algorithm can achieve higher P_D values than other algorithms at all P_F values. For the Salinas dataset, it can be found that, when P_F takes the minimum value, the proposed RTLSR-SP algorithm can obtain the second best detection rate P_D after CRD. In addition, when P_F is greater than 0.05, the ROC curve obtained by the RTLSR-SP algorithm is located in the upper left of other comparison algorithms, which means that, when the segmentation threshold is reduced to this level, the proportion of abnormal targets in the RTLSR-SP algorithm is higher than that of other comparison algorithms. The 2-D ROC curve of (P_D, τ) reflects the target detection ability of a detector, and the closer the curve is to the upper right corner, the stronger the target detection ability is. As shown in Figs. 4(c)–10(c), it can be found that, compared with matrix-based methods (LRX, SegRX, CRD, LRASR, and LSMAD), the proposed RTLSR-SP detector achieves optimal target detection performance on all datasets. Moreover, compared with other detectors, the proposed RTLSR-SP detector obtains optimal or suboptimal target detection performance except for Salinas, San Diego-I, and Hyperion datasets. Regarding the 2-D ROC curve of (P_F, τ), the closer the curve is to the lower left corner, the better the background suppression ability. As can be seen from Figs. 4(d) to 10(d), compared with methods based on tensor representation (PTA, LARTVAD, and PCA-TLRSR), the proposed RTLSR-SP algorithm achieves the optimal background suppression effect on almost all datasets. In addition, compared with other detectors, the background suppression effect of the proposed RTLSR-SP detector on all datasets is at the upper middle level, especially the Pavia Center dataset.

In Fig. 11, black boxes and red boxes denote the detection results corresponding to the background and abnormal components, respectively. As we can see in Fig. 11(a)–(e) and (g), the anomaly background separation performance of the proposed RTLSR-SP is better than that of LRX, SegRX, CRD, LRASR, LSMAD, PTA, and LARTVAD on six datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

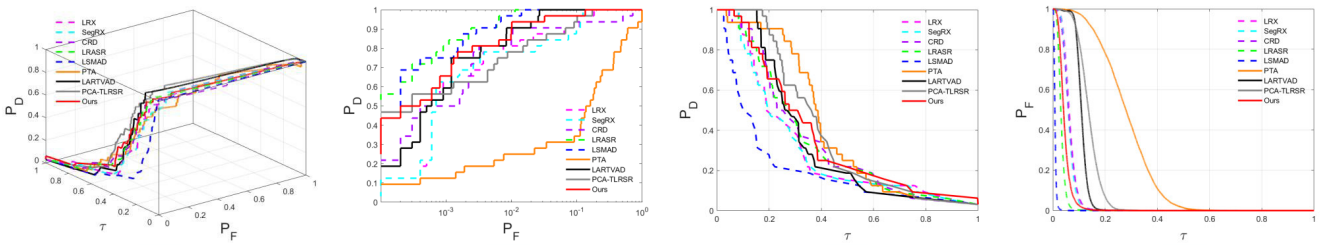


Fig. 7. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the Hyperion dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

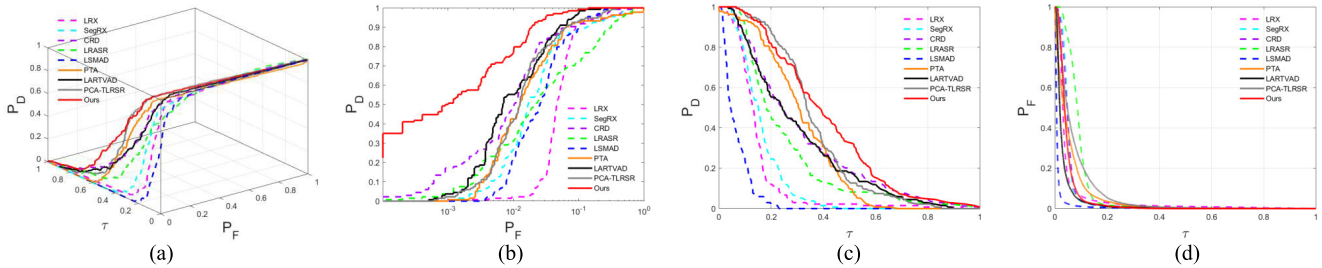


Fig. 8. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the San Diego-II dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

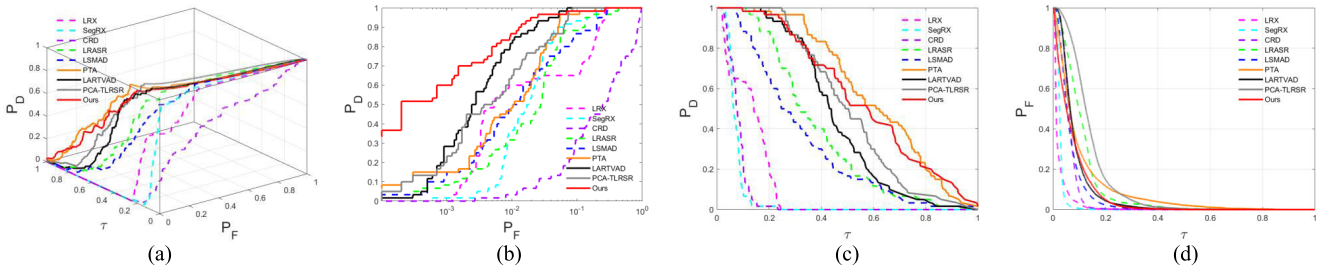


Fig. 9. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the Gulfport dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

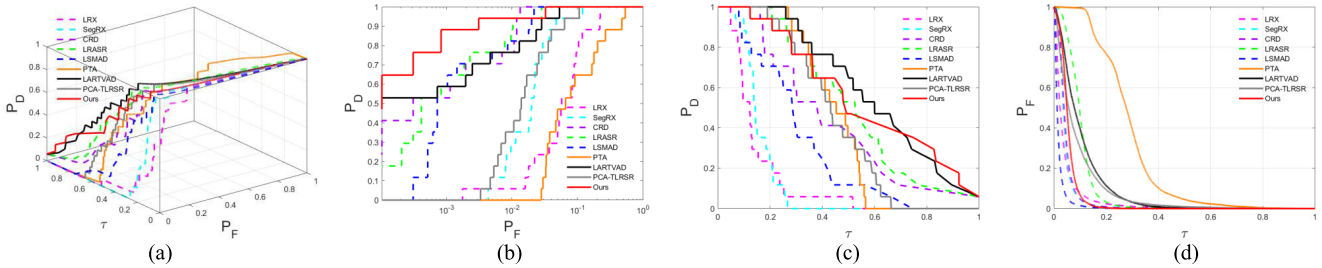


Fig. 10. 3-D and 2-D ROC curves' performance comparison of different anomaly detection approaches on the HYDICE dataset. (a) 3-D ROC. (b) 2-D ROC(P_D, P_F). (c) 2-D ROC(P_D, τ). (d) 2-D ROC(P_F, τ).

Fig. 11(b) shows that PCA-TLRSR reaches the optimal anomaly background separation performance among all detectors on the San Diego-I dataset, and the proposed RTLSR-SP is very close to PCA-TLRSR. As shown in Fig. 11(f), PTA achieves the best anomaly background separation performance on the Gulfport dataset, followed by our proposed RTLSR-SP algorithm. In general, the proposed RTLSR-SP algorithm obtains a good anomaly background separation performance between background and anomalies on almost all datasets.

2) Quantitative Comparison: We report the AUC values of different detectors on seven HSI datasets in Table III, where the optimal and suboptimal results are highlighted in bold and blue underline, respectively. As described in Table III, the proposed RTLSR-SP detector achieves the highest $\text{AUC}_{(D,F)}$ on the Salinas, San Diego-I, San Diego-II, and HYDICE datasets and obtains suboptimal $\text{AUC}_{(D,F)}$ on the Gulfport dataset. When it comes to the overall detection performance index AUC_{ODP}, the proposed RTLSR-SP detector reaches optimal detection results on six datasets and achieves suboptimal detection results on the San Diego-I dataset. These quantitative test results effectively demonstrate the superiority of the proposed RTLSR-SP detector. Furthermore, the running

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

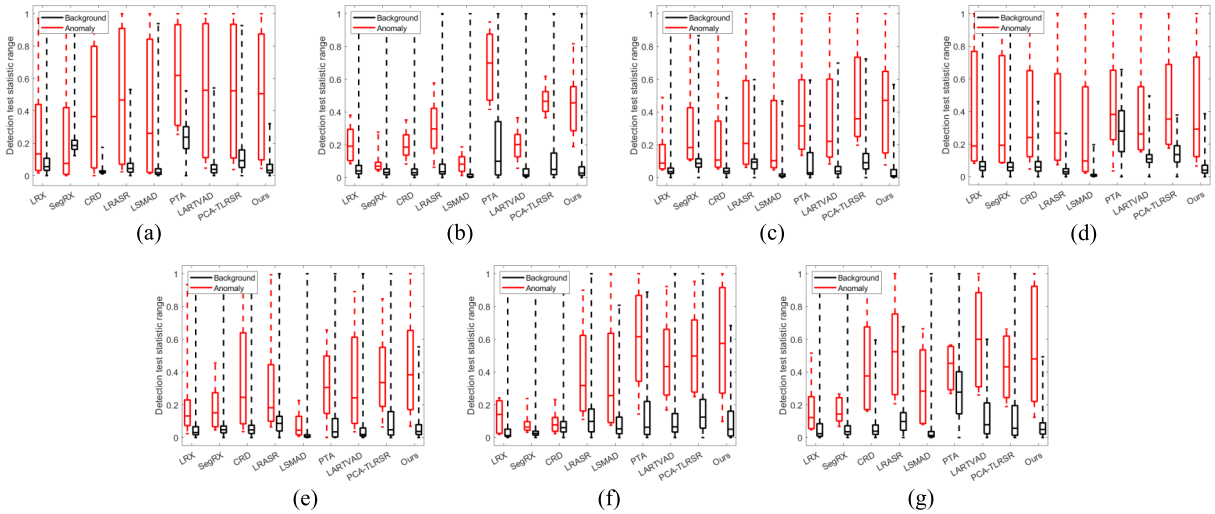


Fig. 11. Anomaly background separation plots of different anomaly detection algorithms on seven HSI datasets. (a) Salinas. (b) San Diego-I. (c) Pavia Center. (d) Hyperion. (e) San Diego-II. (f) Gulfport. (g) HYDICE.

TABLE III

AUC SCORES OF DIFFERENT ANOMALY DETECTION METHODS ON THE SEVEN HSI DATASETS

Datasets	AUC	LRX	SegRX	CRD	LRASR	LSMAD	PTA	LARTVAD	PCA-TLRSR	Ours
Salinas	$AUC_{(D,F)}$	0.7411	0.4711	0.9199	0.9476	0.9207	0.9674	0.9802	0.9228	**0.9844**
	$AUC_{(D,\tau)}$	0.2273	0.1940	0.4495	0.5126	0.3835	**0.6349**	0.5350	0.5527	0.5318
	$AUC_{(F,\tau)}$	0.0687	0.1862	**0.0223**	0.0472	0.0228	0.2453	0.0413	0.1005	0.0388
	AUC_{SNPR}	3.3075	1.0419	**20.1178**	10.8682	16.7985	2.5878	12.9614	5.4986	13.6886
	AUC_{TDBS}	0.1586	0.0078	0.4271	0.4654	0.3607	0.3896	**0.4937**	0.4522	0.4929
	AUC_{ODP}	0.8997	0.4789	1.3470	1.4130	1.2814	1.3569	1.4739	1.3750	**1.4773**
San Diego-I	$AUC_{(D,F)}$	0.9806	0.9418	0.9911	0.9911	0.9844	0.9926	0.9907	0.9957	**0.9986**
	$AUC_{(D,\tau)}$	0.2018	0.0737	0.1928	0.3054	0.0848	**0.6901**	0.2057	0.4686	0.4504
	$AUC_{(F,\tau)}$	0.0486	0.0360	0.0360	0.0475	**0.0113**	0.1416	0.0263	0.0717	0.0367
	AUC_{SNPR}	4.1488	2.0462	5.3570	6.4225	7.5081	4.8744	7.8190	6.5354	**12.2696**
	AUC_{TDBS}	0.1531	0.0377	0.1568	0.2578	0.0735	**0.5485**	0.1794	0.3969	0.4137
	AUC_{ODP}	1.1338	0.9795	1.1479	1.2489	1.0579	**1.5411**	1.1701	1.3925	1.4123
Pavia Center	$AUC_{(D,F)}$	0.9297	0.9685	0.9833	0.7870	**0.9978**	0.9536	0.9950	0.9968	0.9955
	$AUC_{(D,\tau)}$	0.1136	0.2443	0.1829	0.2672	0.2270	0.3593	0.3036	0.4359	**0.4563**
	$AUC_{(F,\tau)}$	0.0479	0.0889	0.0403	0.0905	**0.0127**	0.0573	0.0440	0.0969	0.0211
	AUC_{SNPR}	2.3728	2.7485	4.5386	2.9507	17.9093	6.2664	6.8949	4.4969	**21.6080**
	AUC_{TDBS}	0.0657	0.1554	0.1426	0.1766	0.2143	0.3020	0.2596	0.3390	**0.4352**
	AUC_{ODP}	0.9954	1.1239	1.1260	0.9636	1.2122	1.2555	1.2546	1.3358	**1.4307**
Hyperion	$AUC_{(D,F)}$	0.9830	0.9812	0.9570	**0.9990**	0.9988	0.7472	0.9964	0.9834	0.9933
	$AUC_{(D,\tau)}$	0.3157	0.3100	0.3492	0.3523	0.2218	**0.4190**	0.3480	0.4060	0.3736
	$AUC_{(F,\tau)}$	0.0647	0.0611	0.0625	0.0324	**0.0084**	0.2807	0.1105	0.1385	0.0449
	AUC_{SNPR}	4.8776	5.0655	5.5874	10.8587	**26.3249**	1.4929	3.1487	2.9323	8.3191
	AUC_{TDBS}	0.2510	0.2488	0.2867	0.3199	0.2134	0.1383	0.2375	0.2676	**0.3287**
	AUC_{ODP}	1.2340	1.2300	1.2436	1.3188	1.2122	0.8855	1.2338	1.2509	**1.3220**
San Diego-II	$AUC_{(D,F)}$	0.9281	0.9476	0.9675	0.9052	0.9589	0.9435	0.9797	0.9689	**0.9923**
	$AUC_{(D,\tau)}$	0.1455	0.1610	0.3126	0.2529	0.0651	0.3128	0.2995	0.3633	**0.4088**
	$AUC_{(F,\tau)}$	0.0479	0.0523	0.0527	0.0892	**0.0115**	0.0544	0.0314	0.0726	0.0460
	AUC_{SNPR}	3.0355	3.0772	5.9354	2.8352	5.6783	5.7455	**9.5443**	5.0065	8.8792
	AUC_{TDBS}	0.0976	0.1087	0.2599	0.1637	0.0537	0.2584	0.2682	0.2907	**0.3628**
	AUC_{ODP}	1.0257	1.0562	1.2274	1.0689	1.0125	1.2019	1.2478	1.2596	**1.3551**
Gulfport	$AUC_{(D,F)}$	0.9299	0.9682	0.6473	0.9560	0.9545	0.9788	**0.9921**	0.9840	0.9916
	$AUC_{(D,\tau)}$	0.1281	0.0721	0.0803	0.3805	0.3211	**0.6145**	0.4616	0.5149	0.5774
	$AUC_{(F,\tau)}$	**0.0223**	0.0273	0.0643	0.1093	0.0670	0.1012	0.0820	0.1392	0.0710
	AUC_{SNPR}	5.7548	2.6440	1.2482	3.4818	4.7902	6.0702	5.6275	3.6984	**8.1301**
	AUC_{TDBS}	0.1058	0.0448	0.0160	0.2712	0.2541	**0.5133**	0.3796	0.3756	0.5064
	AUC_{ODP}	1.0357	1.0130	0.6633	1.2272	1.2086	1.4921	1.3717	1.3596	**1.4980**
HYDICE	$AUC_{(D,F)}$	0.9220	0.9746	0.9964	0.9962	0.9952	0.8616	0.9925	0.9765	**0.9978**
	$AUC_{(D,\tau)}$	0.1472	0.1584	0.4449	0.5321	0.3094	0.4404	**0.6217**	0.4366	0.5856
	$AUC_{(F,\tau)}$	0.0408	0.0437	0.0459	0.1005	**0.0190**	0.2832	0.1015	0.0876	0.0542
	AUC_{SNPR}	3.6077	3.6247	9.6911	5.2939	**16.2536**	1.5547	6.1256	4.9816	10.8024
	AUC_{TDBS}	0.1064	0.1147	0.3990	0.4316	0.2904	0.1571	0.5202	0.3490	**0.5314**
	AUC_{ODP}	1.0284	1.0892	1.3954	1.4278	1.2856	1.0187	1.5127	1.3255	**1.5292**

time of all detectors is shown in Table IV, as we can see that the SegRX detector has the shortest running time on most of the datasets, followed by the CRD and LRX detectors. Tensor-based methods (PTA, LARTVAD, PCA-TLRSR, and RTLSR-SP) generally take longer to run than matrix-based methods due to the high computational complexity of tensor

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RUNNING TIME OF DIFFERENT ANOMALY DETECTION METHODS ON SEVEN DATASETS

Datasets	LRX	SegRX	CRD	LRASR	LSMAD	PTA	LARTVAD	PCA-TLRSR	Ours
Salinas	2.9388	1.8029	**1.5538**	17.2259	9.2516	21.3441	37.7454	7.8727	89.5457
San Diego-I	2.9991	**1.1938**	3.9751	9.7149	6.4139	14.3137	17.9651	1.895	59.8642
Pavia Center	1.0632	0.3856	**0.3541**	10.7594	4.4477	9.6267	22.0321	7.3639	32.8688
Hyperion	2.0696	1.6657	**0.7691**	12.3757	4.0757	7.6340	27.8031	6.4431	22.3277
San Diego-II	2.7072	**1.2701**	3.2421	10.007	5.8792	13.8337	26.2059	1.6903	53.2737
Gulfport	2.0612	**1.0195**	2.8703	9.7538	7.4667	14.2212	17.9452	6.2413	56.3324
HYDICE	2.1131	**1.2447**	2.629	13.2054	4.7383	7.117	24.992	6.3807	46.3295

operations. It is worth noting that the PCA-TLRSR method has the shortest running time among the methods based on tensor representation, which may be due to the implementation of the PCA preprocessing step. Although the running time of the proposed RTLSR-SP method is not as good as other comparison methods, the superiority of the RTLSR-SP method can be seen through a large number of qualitative and quantitative experiments. Overall, the running time of the proposed RTLSR-SP approach is acceptable within a certain range.

From these qualitative and quantitative experiments, it can be seen that, among the matrix-based approaches (LRX, SegRX, CRD, LRASR, and LSMAD), the detection performance of the LSMAD method is slightly better, followed by LRASR and CRD. Among the tensor-based methods (PTA, LARTVAD, PCA-TLRSR, and RTLSR-SP), PTA has the worst background suppression effect on the seven datasets, followed by PCA-TLRSR and LARTVAD, which may be because of the proposed RTLSR-SP algorithm makes full use of the prior information of background component and abnormal component. Compared with the PCA-TLRSR algorithm, the proposed RTLSR-SP algorithm achieves the best background suppression effect and target detection performance on most datasets, which is mainly because the RTLSR-SP algorithm captures the global and local spatial–spectral structure information of the background component in the HSI tensor data.

When it comes to background complexity, it can be concluded that background complexity is not the primary factor affecting the detection performance. Pavia Center is a small target detection dataset with a relatively simple background, but the detection results of the algorithms using this dataset are not always optimal. This may be due to the fact that the ground targets in the Pavia Center scene are natural rather than artificial, which reduces the gap between background and anomalous targets and makes anomaly detection more difficult. In terms of the size of abnormal targets, it can be obtained that the majority of algorithms demonstrate superior performance in detecting small targets. The phenomenon can be attributed to two possible factors. One is that the presence of mixing effects (combined effects between the background and abnormal targets) along the edges of large targets can potentially detect a task. Another possibility is that many anomaly detection algorithms are mostly designed for small targets, resulting in poor generalization performance of these algorithms to large targets. In contrast, our proposed RTLSR-SP algorithm can detect both large and small targets simultaneously with good detection performance, which is validated on the Gulfport dataset. In summary, on most datasets, we can conclude that the proposed RTLSR-SP algorithm achieves the most comprehensive and reliable detection performance compared to other algorithms.

F. Parameter Tuning and Analysis

The proposed RTLSR-SP algorithm contains three key parameters, namely, λ, α, and β. λ is an important parameter involved in the dictionary construction process. α and β are two balance parameters. The optimal parameter settings of the RTLSR-SP algorithm are shown in Table II. In this subsection, the parameters of the RTLSR-SP algorithm will be tuned and analyzed.

1) Impact of α and β: In our experiments, the value ranges of parameters α and β are {1e-4, 1e-2, 1e-1, 1e1, 1e2, 1e5} and {1e-5, 1e-4, 1e0, 1e1, 1e2, 1e3}, respectively. To investigate the impact of α and β on HAD performance, we initially set λ as a fixed parameter and subsequently determined the optimal combination of α and β for each HSI dataset individually. As illustrated in Fig. 12, it can be seen that, when α and β are determined in the range of {1e-4, 1e-2, 1e-1} and {1e0, 1e1, 1e2, 1e3}, respectively, the $\text{AUC}_{(D,F)}$ values of RTLSR-SP in the San Diego-I, Hyperion, and San Diego-II datasets are high and stable. Similarly, for the Gulfport and Salinas datasets, when α and β take values within the range of {1e-2, 1e-1, 1e1, 1e2, 1e5} and {1e-5, 1e-4}, respectively, the RTLSR-SP algorithm achieves the highest $\text{AUC}_{(D,F)}$ values. In addition, the highest $\text{AUC}_{(D,F)}$ value on the Pavia Center dataset is obtained when $\alpha = 1e2$ and $\beta = 1e3$. For the HYDICE dataset, the optimal combination of α and β is (1e2, 1e2).

2) Impact of λ: In our experiments, the value range of parameter λ is {0.001, 0.003, 0.005, 0.007, 0.009, 0.01, 0.1}. To study the effect of λ on HAD performance, we first fix α and β and then find the optimal value of λ in each dataset. As exhibited in Fig. 13, we present the optimal value of lambda on each dataset from three aspects: detector effectiveness $\text{AUC}_{(D,F)}$, target detection ability $\text{AUC}_{(D,\tau)}$, and background inhibition ability $\text{AUC}_{(F,\tau)}$. It can be observed that $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$ all achieve the best performance when setting $\lambda = 0.001$ on the Salinas and Pavia Center datasets. Similarly, the ideal value of λ is set at 0.01 for the San Diego-II and HYDICE datasets. For the Hyperion dataset, the best value for λ is 0.003. In consideration of target detection ability $\text{AUC}_{(D,\tau)}$ and background suppression ability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

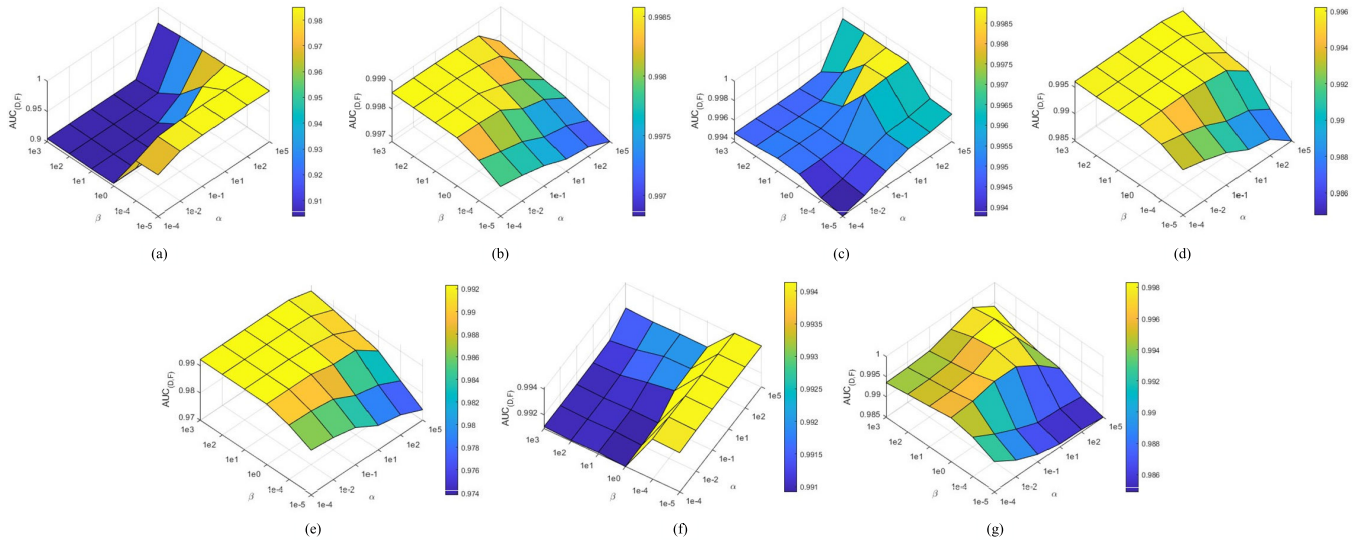


Fig. 12. $\mathrm{AUC}_{(D,F)}$ of the proposed RTLSR-SP algorithm on seven HSI datasets with different α and β combinations. (a) Salinas. (b) San Diego-I. (c) Pavia Center. (d) Hyperion. (e) San Diego-II. (f) Gulfport. (g) HYDICE.

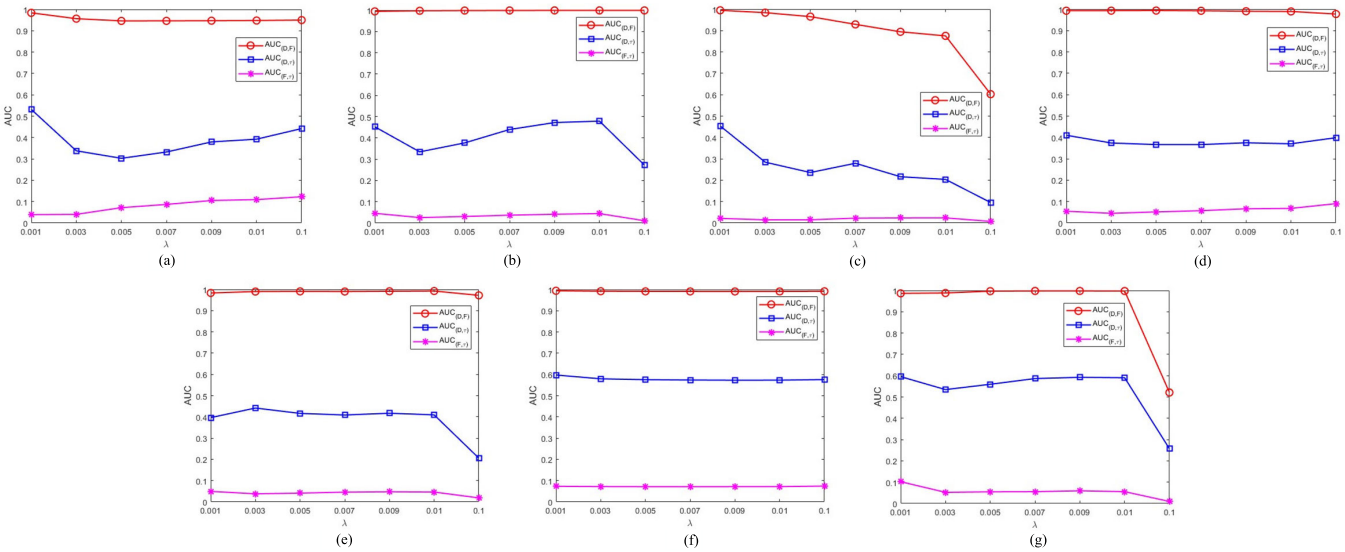


Fig. 13. Curves of $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{(D,\tau)}$, and $\mathrm{AUC}_{(F,\tau)}$ of the RTLSR-SP algorithm on seven HSI datasets with different λ's. (a) Salinas. (b) San Diego-I. (c) Pavia Center. (d) Hyperion. (e) San Diego-II. (f) Gulfport. (g) HYDICE.

TABLE V

$\mathrm{AUC}_{(D,F)}$ PERFORMANCE COMPARISON OF RTLSR-SP ALGORITHM UNDER ORIGINAL HSI AND NOISY HSI

Dataset	Salinas	San Diego-I	Pavia Center	Hyperion	San Diego-II	Gulfport	HYDICE
No noise	**0.9844**	**0.9986**	0.9955	**0.9933**	**0.9923**	**0.9916**	**0.9978**
Gaussian noise (σ=0.001)	**0.9844**	**0.9986**	0.9964	0.9921	0.9921	0.9915	**0.9978**
Gaussian noise (σ=0.005)	0.9824	**0.9986**	**0.9983**	0.9920	0.9904	0.9915	0.9976
Gaussian noise (σ=0.01)	0.9812	0.9984	0.9968	0.9897	0.9914	0.9913	0.9808

$\mathrm{AUC}_{(F,\tau)}$, for San Diego-I and Gulfport datasets, λ is set to 0.007.

In addition, regarding the two tolerate errors ϵ_1 and ϵ_2 in Algorithms 1 and 2, according to experience, we both take the value $1e$-3.

G. Robustness Analysis to Noisy Scenes

Since HSI is inevitably corrupted by noise during data acquisition and transmission, it becomes more challenging to accurately identify anomalous targets on the ground. In order to verify the target detection performance and robustness of the proposed RTLSR-SP algorithm in different noise scenarios, we construct three sets of HSI datasets with different noise intensities by adding noise to each band of each hyperspectral dataset. The added noise is Gaussian noise with zero mean and variances of 0.001, 0.005, and 0.01, respectively. The noise images are shown in Fig. 14(c), (e), and (g), respectively, and the corresponding detection images are displayed in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

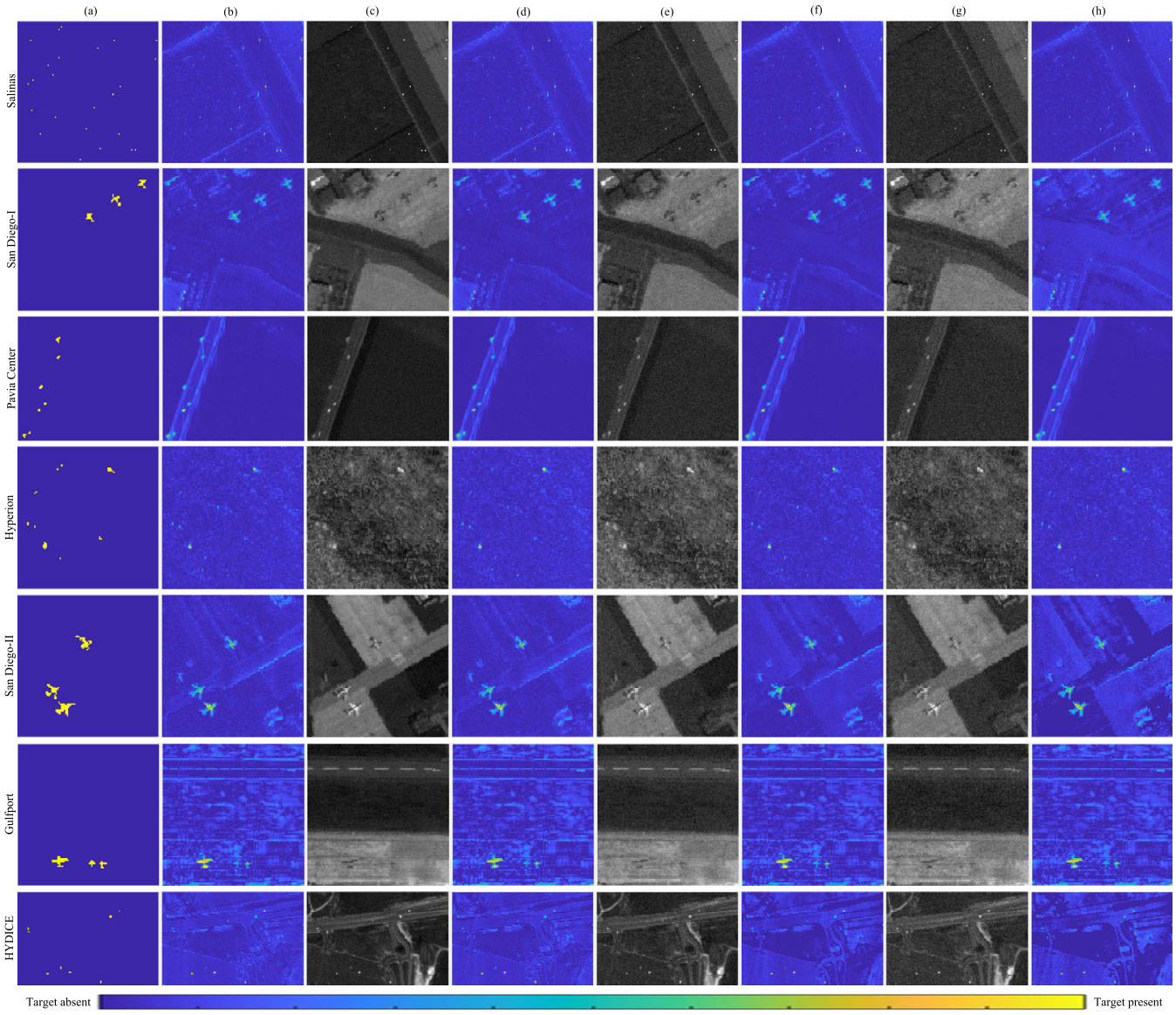


Fig. 14. Detection maps acquired by the proposed algorithm on the original HSI and the noise-added HSI. (a) Ground truth. (b) Detection maps on raw HSI. (c) Gaussian noise dataset with a variance of 0.001. (d) Corresponding detection results in the case of (c). (e) Gaussian noise dataset with a variance of 0.005. (f) Corresponding detection results in the case of (e). (g) Gaussian noise dataset with a variance of 0.01. (h) Corresponding detection results in the case of (g).

Fig. 14(d), (f), and (h). Meanwhile, Table V also lists the $\mathrm{AUC}_{(D,F)}$ values of each HSI dataset.

It can be observed from Fig. 14 that, except for the Pavia Center dataset, the anomaly detection maps of the RTLSR-SP algorithm on the noise-free HSI datasets are slightly better than those on the noisy HSI datasets, which is consistent with the numerical results in Table V. From Fig. 14(d), (f), and (h), it can be found that, as the variance σ increases, the background suppression effect of the RTLSR-SP algorithm on the San Diego-I, San Diego-II, and HYDICE datasets gradually becomes worse. In addition, as shown in Table V, on the Pavia Center dataset, the $\mathrm{AUC}_{(D,F)}$ value of the RTLSR-SP algorithm on noise-free HSI is lower than that on noisy HSI, which is consistent with the visual anomaly detection map in Fig. 14. On other datasets, the $\mathrm{AUC}_{(D,F)}$ values of the RTLSR-SP algorithm on noise-free HSIs are slightly higher than those on noisy HSIs. Although the detection results of the RTLSR-SP algorithm on the noisy HSI datasets are slightly worse, it can still detect abnormal targets effectively. Overall, the proposed RTLSR-SP algorithm is robust against noisy backgrounds.

H. Ablation Study

In this section, we conduct ablation experiments on seven HSI datasets to test the robust dictionary construction method TRPCA, sparse constraint $\|\mathcal{Z}\|_1$ term, skinny t-SVD, and sparse saliency prior weight tensor $\mathcal{W}$ against the proposed RTLSR-SP algorithm detection performance impact. Figs. 15 and 16, respectively, show the comparison results of $\mathrm{AUC}_{(D,F)}$ values with or without TRPCA module and with or without sparse constraint $\|\mathcal{Z}\|_1$ term. The experimental results with or without skinny t-SVD and with or without

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

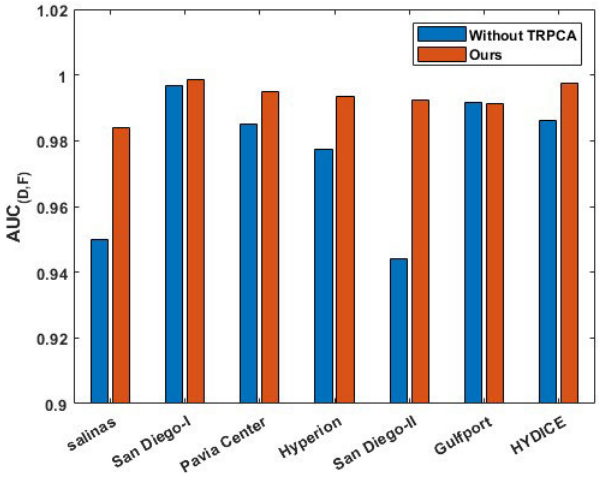


Fig. 15. Comparison of $\mathrm{AUC}_{(D,F)}$ values between RTLSR-SP and RTLSR-SP without the TRPCA module.

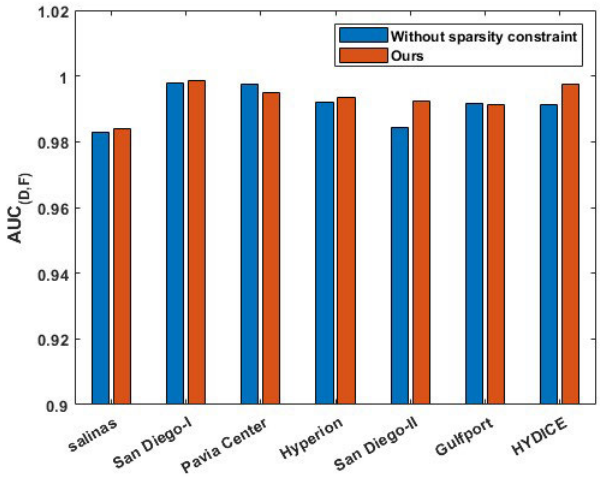


Fig. 16. Comparison of $\mathrm{AUC}_{(D,F)}$ values between RTLSR-SP and RTLSR-SP without the sparse constraint $\|\mathcal{Z}\|_1$ term.

$\mathcal{W}$ are shown in Table VI, where $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{(D,\tau)}$, $\mathrm{AUC}_{(F,\tau)}$, and time (s) are recorded. It is worth noting that, in Table VI, RTLSR-SP without skinny t-SVD and RTLSR-SP without $\mathcal{W}$, respectively, represent the two algorithms of our proposed RTLSR-SP algorithm under the absence of skinny t-SVD or sparse saliency prior weight tensor $\mathcal{W}$. RTLSR-SP without skinny t-SVD and $\mathcal{W}$ is the baseline of our proposed RTLSR-SP algorithm; that is, both skinny t-SVD and sparse saliency prior weight tensor $\mathcal{W}$ are missing. Furthermore, the increased and decreased amounts compared to the full RTLSR-SP algorithm are highlighted in red and blue, respectively.

1) With and Without TRPCA: To investigate the impact of the TRPCA module on detection performance, we test the $\mathrm{AUC}_{(D,F)}$ values of RTLSR-SP with and without the TRPCA step, as shown in Fig. 15. It can be observed that, except for the Gulfport dataset, the $\mathrm{AUC}_{(D,F)}$ values of RTLSR-SP are higher than that of RTLSR-SP without TRPCA, especially on the Salinas and San Diego-II datasets. The reason for this may be that the original hyperspectral data contain noise and outliers, directly using the original data as a dictionary may weaken the performance of anomaly detection. It is obvious that the TRPCA method can filter noise and outliers and improve the detection performance.

2) With and Without Sparse Constraint: To verify the effectiveness of imposing sparsity constraint on coefficient tensor $\mathcal{Z}$, we test the $\mathrm{AUC}_{(D,F)}$ values of the RTLSR-SP algorithm with and without the $\|\mathcal{Z}\|_1$ term. The experimental results are displayed in Fig. 16. It can be easily seen that, compared with the RTLSR-SP algorithm without the $\|\mathcal{Z}\|_1$ term, the RTLSR-SP algorithm achieves the best $\mathrm{AUC}_{(D,F)}$ values except for the Pavia Center and Gulfport datasets. This also verifies that the addition of the sparse constraint $\|\mathcal{Z}\|_1$ term can capture the local structure information of the HSI tensor data and further improve the detection performance.

3) With and Without Skinny t-SVD: In our proposed RTLSR-SP algorithm, we use skinny t-SVD to obtain a more effective background dictionary $\mathcal{D}$. To confirm this point, we carry out an ablation study on skinny t-SVD in Table VI. It can be easily found that the values of $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(F,\tau)}$ fluctuate slightly between the RTLSR-SP algorithm without skinny t-SVD and the RTLSR-SP algorithm using skinny t-SVD. In terms of $\mathrm{AUC}_{(D,\tau)}$ value, the proposed RTLSR-SP algorithm is superior to the RTLSR-SP algorithm without skinny t-SVD on the San Diego-I, Hyperion, Gulfport, and HYDICE datasets, especially the Gulfport dataset, whose value is higher than 10.99%. In addition, in terms of running time, compared with the RTLSR-SP algorithm without skinny t-SVD, the proposed RTLSR-SP algorithm is at the lowest level of the seven datasets, which verifies that using skinny t-SVD can effectively improve the efficiency of the RTLSR-SP algorithm.

4) With and Without Sparse Saliency Prior Weight Tensor: In Section III-B, we design a sparse saliency prior weight map $\boldsymbol{W}$ and then obtain a sparse saliency prior weight tensor $\mathcal{W}$. To explore the importance of $\mathcal{W}$ for the proposed RTLSR-SP algorithm, Table VI reports the detection performance of RTLSR-SP with and without $\mathcal{W}$. It can be observed from Table VI that, except for the Pavia Center and HYDICE datasets, the $\mathrm{AUC}_{(F,\tau)}$ values of the proposed RTLSR-SP algorithm are lower than those of the RTLSR-SP algorithm that does not use $\mathcal{W}$. Moreover, in terms of $\mathrm{AUC}_{(D,\tau)}$, the proposed RTLSR-SP algorithm achieves the best performance compared to the RTLSR-SP algorithm without $\mathcal{W}$ except for the San Diego-I and Gulfport datasets. Similarly, except for the Pavia Center and Gulfport datasets, the $\mathrm{AUC}_{(D,F)}$ values of the RTLSR-SP algorithm that does not use $\mathcal{W}$ are lower than our proposed RTLSR-SP algorithm. Therefore, we can conclude that the design of the saliency prior weight map $\boldsymbol{W}$ is able to enhance the saliency of abnormal targets while suppressing the background.

5) With and Without Skinny t-SVD and Sparse Saliency Prior Weight Tensor: Compared with Table III, it can be found that the three evaluation index values of the RTLSR-SP algorithm that does not use skinny t-SVD and $\mathcal{W}$ at the same time still exceed some competing algorithms, such as LRX, which means that the proposed algorithm framework is effective. From the experimental results in Table VI, it is not difficult to observe that the proposed RTLSR-SP algorithm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF THE PROPOSED RTLSR-SP ALGORITHM ON SEVEN HSI DATASETS

Datasets	Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	Time
salinas	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9848(+0.0004)	0.5314(-0.0004)	0.0383(-0.0005)	92.2274(+2.6817)
	RTLSR-SP without skinny t-SVD	0.9849(+0.0005)	0.5319(+0.0001)	0.0383(-0.0005)	92.5255(+2.9798)
	RTLSR-SP without $\mathcal{W}$	0.9843(-0.0001)	0.5312(-0.0006)	0.0389(+0.0001)	89.96(+0.4143)
	RTLSR-SP	0.9844	0.5318	0.0388	89.5457
San Diego-I	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9976(-0.001)	0.3711(-0.0793)	0.0381(+0.0014)	38.6481(-21.2161)
	RTLSR-SP without skinny t-SVD	0.9985(-0.0001)	0.4316(-0.0188)	0.0299(-0.0068)	60.7863(+0.9221)
	RTLSR-SP without $\mathcal{W}$	0.9983(-0.003)	0.4714(+0.021)	0.0554(+0.0187)	35.7383(-24.1259)
	RTLSR-SP	0.9986	0.4504	0.0367	59.8642
Pavia Center	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9974(+0.0019)	0.4198(-0.0365)	0.0194(-0.0017)	31.7127(-1.1561)
	RTLSR-SP without skinny t-SVD	0.9964(+0.0009)	0.4610(+0.0047)	0.0210(-0.0001)	43.5223(+10.6535)
	RTLSR-SP without $\mathcal{W}$	0.9970(+0.0015)	0.4087(-0.0476)	0.0187(-0.0024)	25.1986(-7.6702)
	RTLSR-SP	0.9955	0.4563	0.0211	32.8688
Hyperion	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9913(-0.002)	0.3617(-0.0119)	0.0492(+0.0043)	25.427(+3.0993)
	RTLSR-SP without skinny t-SVD	0.9939(+0.0006)	0.3669(-0.0067)	0.0423(-0.0026)	24.8945(+2.5668)
	RTLSR-SP without $\mathcal{W}$	0.9904(-0.0029)	0.3668(-0.0068)	0.0516(+0.0067)	22.0806(-0.2471)
	RTLSR-SP	0.9933	0.3736	0.0449	22.3277
San Diego-II	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9880(-0.0043)	0.4025(-0.0063)	0.0474(+0.0014)	31.8341(-21.4396)
	RTLSR-SP without skinny t-SVD	0.9918(-0.0005)	0.4123(+0.0035)	0.0532(+0.0072)	53.8295(+0.5558)
	RTLSR-SP without $\mathcal{W}$	0.9883(-0.004)	0.4051(-0.0037)	0.0537(+0.0077)	29.5886(-23.6851)
	RTLSR-SP	0.9923	0.4088	0.0460	53.2737
Gulfport	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9880(-0.0036)	0.3869(-0.1905)	0.0788(+0.0078)	44.152(-12.1804)
	RTLSR-SP without skinny t-SVD	0.9894(-0.0022)	0.4675(-0.1099)	0.0845(+0.0135)	60.417(+4.0846)
	RTLSR-SP without $\mathcal{W}$	0.9944(+0.0028)	0.5844(+0.007)	0.0711(+0.0001)	35.5713(-20.7611)
	RTLSR-SP	0.9916	0.5774	0.0710	56.3324
HYDICE	RTLSR-SP without skinny t-SVD and $\mathcal{W}$	0.9975(-0.0003)	0.5728(-0.0128)	0.0449(-0.0093)	49.5822(+3.2527)
	RTLSR-SP without skinny t-SVD	0.9980(+0.0002)	0.5733(-0.0123)	0.0498(-0.0044)	67.2976(+20.9681)
	RTLSR-SP without $\mathcal{W}$	0.9971(-0.0007)	0.5749(-0.0107)	0.0455(-0.0087)	33.3225(-13.007)
	RTLSR-SP	0.9978	0.5856	0.0542	46.3295

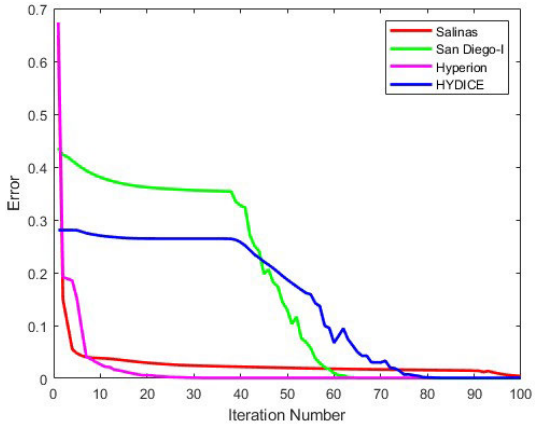


Fig. 17. Convergence curves of the proposed RTLSR-SP algorithm on four datasets.

achieves the best detection performance compared with the RTLSR-SP algorithm that does not use skinny t-SVD and $\mathcal{W}$ at the same time except for the Salinas and Pavia Center datasets, which illustrates the importance of jointly using skinny t-SVD and sparse saliency prior weight tensor.

I. Convergence and Complexity Analysis

1) Convergence Analysis: Fig. 17 demonstrates the convergence error curves of the proposed RTLSR-SP algorithm on the Salinas, San Diego-I, Hyperion, and HYDICE datasets as the number of iterations changes. The error is represented by the convergence condition (25). It is easy to see from Fig. 17 that, as the number of iterations increases, the error curves on the four datasets continue to decrease until the convergence condition of the algorithm is reached. Thereby, the proposed RTLSR-SP algorithm has good convergence in HAD.

2) Complexity Analysis: The primary computational burden in our proposed RTLSR-SP algorithm arises from the iterative updating of $\mathcal{Z}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{E}$. In each iteration, the update of $\mathcal{Z}$ in problem (14) consists of tensor t-products and tensor inversions, with a total computational cost of $\mathcal{O}(r_{\mathcal{L}}(h+w)b\log(b)+r_{\mathcal{L}}hwb)$. Regarding the update of $\mathcal{B}$, the cost is $\mathcal{O}(r_{\mathcal{L}}^2wb)$. The update about $\mathcal{C}$ in problem (19) costs $\mathcal{O}(r_{\mathcal{L}}wb)$. Updating $\mathcal{E}$ costs $\mathcal{O}(r_{\mathcal{L}}hwb)$. Therefore, the total computational complexity consists of the sum of these four parts, where t represents the number of iterations required for the proposed RTLSR-SP algorithm to converge.

V. CONCLUSION

In this article, a novel TLRR-based RTLSR-SP model for the HAD task is proposed. Considering the low-rank and local structural similarity of the background, we impose both low-rank and sparse constraints on the coefficient tensor to capture the global and local spatial–spectral structure information of the HSI tensor data, respectively. Furthermore, a sparse saliency prior weight tensor is devised for the anomaly tensor to enhance the saliency of anomalous targets. Meanwhile, the $\ell_{F,1}$-norm is also incorporated into our model to promote group sparsity of anomalous targets. Next, a robust and efficient background dictionary is developed, which is robust to noise and outliers, and adequately represents background pixels. Finally, an efficient iterative algorithm based on ADMM is proposed to optimize the RTLSR-SP model. Extensive experimental findings on simulated and real HSI datasets show that the proposed RTLSR-SP approach is more effective and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

superior to current state-of-the-art HAD approaches. Since dictionary learning and detection models are performed in two steps, this inevitably leads to suboptimal solutions. The future focus of our research will revolve around the integration of dictionary learning and anomaly detection models within a unified framework, aiming to enhance the performance of anomaly detection.

REFERENCES

[1] S. Wang et al., "Subfeature ensemble-based hyperspectral anomaly detection algorithm," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5943–5952, 2022.

[2] S. Yuan, L. Shi, B. Yao, F. Li, and Y. Du, "A hyperspectral anomaly detection algorithm using sub-features grouping and binary accumulation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[3] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[4] Z. Xue, X. Nie, and M. Zhang, "Incremental dictionary learning-driven tensor low-rank and sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544019.

[5] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[6] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[7] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[8] N. Gorelnik, H. Yehudai, and S. R. Rotman, "Anomaly detection in non-stationary backgrounds," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[9] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[10] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[11] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[12] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[13] S. Feng, S. Tang, C. Zhao, and Y. Cui, "A hyperspectral anomaly detection method based on low-rank and sparse decomposition with density peak guided collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501513.

[14] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[15] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Joint reconstruction and anomaly detection from compressive hyperspectral images using Mahalanobis distance-regularized tensor RPCA," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2919–2930, May 2018.

[16] S. Feng, D. Wu, R. Feng, and C. Zhao, "Hyperspectral anomaly detection with total variation regularized low rank tensor decomposition and collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[17] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

[18] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[19] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[20] R. Feng, H. Li, L. Wang, Y. Zhong, L. Zhang, and T. Zeng, "Local spatial constraint and total variation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512216.

[21] C. Zhao, C. Li, S. Feng, and X. Jia, "Enhanced total variation regularized representation model with endmember background dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518312.

[22] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[23] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[24] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[25] Z. Hou, W. Li, R. Tao, P. Ma, and W. Shi, "Collaborative representation with background purification and saliency weight for hyperspectral anomaly detection," *Sci. China Inf. Sci.*, vol. 65, no. 1, Jan. 2022, Art. no. 112305.

[26] G. Zhang, N. Li, B. Tu, Z. Liao, and Y. Peng, "Hyperspectral anomaly detection via dual collaborative representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4881–4894, 2020.

[27] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[28] S. Lin, M. Zhang, X. Cheng, L. Wang, M. Xu, and H. Wang, "Hyperspectral anomaly detection via dual dictionaries construction guided by two-stage complementary decision," *Remote Sens.*, vol. 14, no. 8, p. 1784, Apr. 2022.

[29] T. Cheng and B. Wang, "Total variation and sparsity regularized decomposition model with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1472–1486, Feb. 2021.

[30] Z. Wu and B. Wang, "Kernel-based decomposition model with total variation and sparsity regularizations via union dictionary for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542916.

[31] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[32] X. He, J. Wu, Q. Ling, Z. Li, Z. Lin, and S. Zhou, "Anomaly detection for hyperspectral imagery via tensor low-rank approximation with multiple subspace learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509917.

[33] J. Wang, J. Sun, Y. Xia, and Y. Zhang, "Hyperspectral anomaly detection via weighted-sparsity-regularized tensor linear representation," *IET Image Process.*, vol. 17, no. 4, pp. 1029–1043, Mar. 2023.

[34] S. Sun, J. Liu, X. Chen, W. Li, and H. Li, "Hyperspectral anomaly detection with tensor average rank and piecewise smoothness constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8679–8692, Nov. 2023.

[35] S. Sun, J. Liu, and W. Li, "Spatial invariant tensor self-representation model for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, early access, Jan. 18, 2023, doi: 10.1109/TCYB.2022.3233108.

[36] Z. Li, Y. Zhang, and J. Zhang, "Tensor approximation with low-rank representation and kurtosis correlation constraint for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533713.

[37] A. Ruhan, X. Mu, and J. He, "Enhance tensor RPCA-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[38] Y.-P. Zhao, H. Li, Y. Chen, Z. Wang, and X. Li, "Hyperspectral anomaly detection via structured sparsity plus enhanced low-rankness," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515115.

[39] Y. Wang, D. Li, H. Wu, X. Li, F. Kong, and Q. Wang, "Multiple spectral–spatial representation based on tensor decomposition for HSI anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3539–3551, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.

[40] M. Feng, W. Chen, Y. Yang, Q. Shu, H. Li, and Y. Huang, "Hyperspectral anomaly detection based on tensor ring decomposition with factors TV regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514114.
[41] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis with a new tensor nuclear norm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 925–938, Apr. 2020.
[42] J. Huang, K. Liu, and X. Li, "Locality constrained low rank representation and automatic dictionary learning for hyperspectral anomaly detection," *Remote Sens.*, vol. 14, no. 6, p. 1327, Mar. 2022.
[43] S. Du, Y. Shi, G. Shan, W. Wang, and Y. Ma, "Tensor low-rank sparse representation for tensor subspace learning," *Neurocomputing*, vol. 440, pp. 351–364, Jun. 2021.
[44] P. Zhou, C. Lu, J. Feng, Z. Lin, and S. Yan, "Tensor low-rank representation for data recovery and clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 5, pp. 1718–1732, May 2021.
[45] Y. Luo, X. Li, S. Chen, C. Xia, and L. Zhao, "IMNN-LWEC: A novel infrared small target detection based on spatial–temporal tensor model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5004022.
[46] R. Wang, H. Hu, F. He, F. Nie, S. Cai, and Z. Ming, "Self-weighted collaborative representation for hyperspectral anomaly detection," *Signal Process.*, vol. 177, Dec. 2020, Art. no. 107718.
[47] X. Shen, H. Liu, J. Nie, and X. Zhou, "Matrix factorization with framelet and saliency priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504413.
[48] S. Chang and P. Ghamisi, "Nonnegative-constrained joint collaborative representation with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534913.
[49] G. Liu, Z. Lin, and Y. Yu, "Robust subspace segmentation by low-rank representation," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 663–670.
[50] M. Yin, J. Gao, S. Xie, and Y. Guo, "Multiview subspace clustering via tensorial t-product representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 3, pp. 851–864, Mar. 2019.
[51] P. Xiang et al., "Hyperspectral anomaly detection by local joint subspace process and support vector machine," *Int. J. Remote Sens.*, vol. 41, no. 10, pp. 3798–3819, May 2020.
[52] E. J. Candes, M. B. Wakin, and S. P. Boyd, "Enhancing sparsity by reweighted $\ell 1$ minimization," *J. Fourier Anal. Appl.*, vol. 14, nos. 5–6, pp. 877–905, 2008.
[53] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.
[54] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.
[55] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.



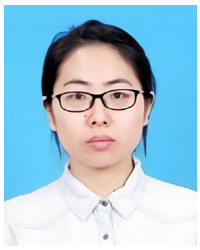
Qingjiang Xiao (Student Member, IEEE) received the M.S. degree in computer technology from Northwest Minzu University, Lanzhou, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Computer Science, Hangzhou Dianzi University, Hangzhou, China.

His research interests include hyperspectral image processing.



Liaoying Zhao received the B.S. and M.S. degrees from Hangzhou Dianzi University, Hangzhou, China, in 1992 and 1995, respectively, and the Ph.D. degree from Zhejiang University, Hangzhou, in 2004.

Since 1995, she has been with Hangzhou Dianzi University, where she is currently a Professor with the College of Computer Science. Her research interests include hyperspectral image processing, signal and image processing, pattern recognition, and machine learning.



Shuhan Chen (Member, IEEE) received the B.S. degree from Ludong University, Yantai, China, in 2011, the M.S. degree from Liaoning Technical University, Huludao, China, in 2014, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2020.

She did research as a Visiting Faculty Assistant at the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, from 2018 to 2020. She currently holds a post-doctoral position at the Department of Control Science and Engineering, Zhejiang University. Her research interests include image registration, hyperspectral image processing, and pattern recognition.



Xiaorun Li received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1995 and 2008, respectively.

Since 1995, he has been with Zhejiang University, where he is currently a Professor with the College of Electrical Engineering. His research interests include hyperspectral image processing, signal and image processing, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:42:00 UTC from IEEE Xplore. Restrictions apply.